TECHNICAL REFERENCE / EDITION 3 / 7 SEPTEMBER 2026

Full 600-cell 35-orbit algorithm reference

Target seeds, special seeds, star operations, parity and verified solve records

A star operation is one setup-conjugated three-piece cycle. Its target-orbit action cycles two designated buffer positions A, B and a chosen target position X. It is one macro application, not one primitive turn. The two buffers are held fixed during the setup search; their occupants move during the star itself.

$$K_{o,X,q} = S_{X,q}^{-1} R_o^{-1} W_o R_o S_{X,q}, \qquad K_{o,X,q}|_o = (A\ B\ X).$$

This reference translates the retained toolkit into explicit formulas and one algorithm card for every moving orbit. All 35 formulas have been reduced and compared to the exact stored words. Their complete sticker effects have been recalculated. The three random full-solve traces, the lifted-log cleanup trace, and the older pure-controller trace have been replayed from their recorded inputs. All finish with every one of the 259,800 labelled stickers at home. These are computational results for the retained model. [S3,S4]

35 moving orbits	**177,120 pieces**	**259,800 stickers**	**12 pure raw seeds**
600 further fixed singleton orbits	176,520 movable pieces; 600 fixed centers	433 sticker regions in each tetrahedral cell	23 raw seeds have explicitly listed collateral

Execution boundary. No Windows GUI solve is being claimed. The supplied full native-format exports are complete expanded archival move sequences, with independently checked syntax, counts and checksums. Each expands to roughly 2.9 GB. The original MPUlt history loader grows a single 64-bit-entry array and is unsuitable for loading these traces. Use the compressed replay package for verification. A full native geometry bridge and GUI replay remain untested. [S4,S6]

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1. Notation and move addresses

Three different numbering systems

Orbit IDs O00 through O34 identify move-connected piece families in the retained model. **Cell IDs** $c = 0, \ldots, 599$ identify tetrahedral cells. **Piece and sticker IDs** identify positions in the lab model; they are not native MPUlt menu labels. Numeric buffer IDs on an orbit card are piece-position IDs.

A native twist token is written `axis:twist:angle:mask`. The native axis identifies one of 300 antipodal pairs, while mask 1 or 4 selects its positive or negative outer cap; mask 5 selects both. The fixed middle mask bit is 2. Native indices were reconstructed in source expansion order and checked by replaying the supplied simplified log. The cards give native tokens for every primitive they require. [S1,S2,S5,S6]

The two lab primitives at cell c

H_c is the specified order-two cell-cap rotation; T_c is the specified order-three rotation. Thus $H_c^{-1} = H_c$ and $T_c^{-1} = T_c^2$. Their positive lab integer codes are

$$H_c \longleftrightarrow 2c + 1, \qquad T_c \longleftrightarrow 2c + 2.$$

A negative code denotes the inverse. A native angle 2 for an order-three twist is one inverse-turn record, not two stored records. Every lab primitive in this reference has a single native-token translation. The named H_c, T_c depend on the fixed lab cell frame, not on the camera direction.

Reading the formulas

Words execute **from left to right**. $[P, Q] = PQP^{-1}Q^{-1}$ is a commutator. Inverting a word reverses its order and inverts every move. The compact left-nested chain is

$$\mathcal{C}(P; g_1, \ldots, g_m) = [\cdots[[P, g_1], g_2], \ldots, g_m].$$

For example, $\mathcal{C}(H_0; T_1, H_{15}) = [[H_0, T_1], H_{15}]$. A power repeats the *whole* enclosed word. Three shared helper words shorten the cards:

$$\boxed{U = [H_0, T_1], \qquad V = [H_0, T_5], \qquad Z = [[U^4, T_{18}], H_0].}$$

A reference such as $[W_{09}, H_{28}]$ uses the entire recorded seed W_{09}, including all its collateral effects. It never means substituting only its idealized three-cycle.

Count convention. Card lengths are the exact stored, reduced words. Some displayed commutator expressions expand to longer equivalent words before adjacent same-generator cancellation. Apply $H_c^2 = 1$ and $T_c^3 = 1$ to recover the stored length. All 35 normal-form comparisons passed. The exact forward and inverse bodies in both notations are supplied as `algorithms/Oxx_exact_words.txt`; no moves are omitted. [S4]

2. What a star operation actually does

A target seed W_o cycles precisely three pieces *within orbit* o. It can also move pieces in other orbits. A short relocation R_o places its first two targets at the certified antipodal buffers. Set

$$K_o = R_o^{-1} W_o R_o.$$

On the target orbit this is $(A\ B\ C)$, with specified sticker transport. The setup $S_{X,q}$ carries the reference third piece C to destination X in orientation frame q, while fixing both buffers. Executing S^{-1} first brings X into the seed's reference position. Hence

$$\boxed{K_{o,X,q} = S_{X,q}^{-1} K_o S_{X,q}}$$

is one forward star. Its inverse is one inverse star. Both count as one operation in the saved command traces. All stars for the orbit share the same pair of buffers, which is the reason for the name used by this toolkit. [S3,S4]

	Buffer A	Buffer B	Target X
Before	α	β	χ
After forward $(A\ B\ X)$	χ	α	β
After inverse $(A\ X\ B)$	β	χ	α

How the setup and orientation are specified

A physical position has a cap-membership set $M(P)$ and a hosting-cell set $H(P)$. A primitive of cap c moves that piece exactly when $c \in M(P)$. Exclude every cap in $M(A) \cup M(B)$ from the setup search. Search states are *ordered sticker frames at positions*, so a node records both X and q.

The stored breadth-first trees use positive H_c, T_c edges. Each reaches exactly $(N_o - 2)|\Omega_o|$ states, covering every nonbuffer position in every attainable orientation. "Shortest setup" refers to this positive-edge, buffer-protecting search, not to global optimality in the full native move alphabet. The trees, parent moves and ordered frames are included in `execution_trees.json.gz`. [S3,S4]

How many primitive turns is one star?

If the stored seed has length ℓ_o, relocation length r_o, and setup length d, its recorded expansion has

$$\boxed{L_{\text{star}}(o, X, q) = \ell_o + 2r_o + 2d.}$$

The same count applies to its inverse. Across the certified controllers and setup trees, this ranges from 56 to 12,304 primitives. In the recorded seed-600 solve it averages 942.595. One star is therefore a useful macro-level workload unit, but a very poor substitute for a primitive-turn count.

> The buffers stay fixed *during the setup*; all three occupants move *during the star*. A raw star can disturb its listed collateral orbits. The identity $K_o^3 = 1$ is guaranteed on the target orbit, not automatically on the whole puzzle. Keep the actual full word and full net permutation. [S3,S4]

3. The orbit-solving procedure

Place the pieces

Keep $\mathrm{where}(p)$, the current position of the piece whose home is p. Visit each destination X outside the buffers. If its correct piece is already there, continue. Otherwise put $Y = \mathrm{where}(X)$ and use the canonical frame star K_X:

Required piece is at	Execute	Reason
A	K_X^{-1}	The inverse sends A to X.
B	K_X	The forward star sends B to X.
Another position Y	$K_Y^{-1} K_X$	The net target-orbit cycle is $(A\ Y\ X)$; Y goes to X.

Every insertion preserves already completed nonbuffer positions in the target orbit. Once they are all correct, even positional parity forces the two remaining buffer pieces to be correct too. This conclusion applies to tracked legal identities. [S3,S4]

Orient the pieces

Position completion need not orient multi-sticker pieces. Let $K_{X,q}$ use frame q at X, and K_X use its canonical frame. The two-star word

$$\boxed{P_X(q) = K_{X,q} K_X^{-1}}$$

fixes positions and transfers inverse orientation changes between B and X. Enumerate the actual local frames and choose the one that puts every sticker of X at home. Its full orientation space has 1, 2, 5, 10 or 60 elements, depending on the orbit. Apply this to each nonbuffer piece requiring orientation.

To orient A, use the conjugated transfer $K_o P_C(q) K_o^{-1}$. The implementation uses its three-star target-equivalent form

$$\boxed{K_o K_{C,q} K_o.}$$

In raw triangular mode the full collateral of those three actual words is retained. This substitution is not permission to discard collateral by algebra done only on the target orbit.

For the last buffer B, choose distinct nonbuffers X, Y and use

$$\boxed{[P_X(q), P_Y(r)]}$$

when needed. This is eight stars and changes only B on the target orbit. The pair (q, r) is chosen from the actual finite orientation group to produce the required residual. The cyclic orbits need no such step because their final residual is forced to be identity. The nonabelian cases are explained next. [S3,S4]

Protect the rest of the puzzle

Use the exact triangular order on page 6. A seed's collateral is confined to unfinished orbits in that order. Apply complete sticker permutations, not just the target three-cycle; check every completed orbit after each stage. Pure controllers allow other orders by adding certified collateral corrections, but their current primitive words are much longer. Visibility filters have no bearing on what is preserved.

4. Parity and orientation endgames

The reported invariants are specific to the retained full-cut model and its legal generators. The existing exhaustive audit checks all 1,200 positive primary generators; inverses and compositions inherit the zero invariants. The identities below use the toolkit's transported orientation frames. [S3,S4]

Family	Constraint on a legal state	Consequence after all other pieces in that orbit are solved
Every moving orbit	$\mathrm{sgn}(\pi_o) = +1$	Exactly two swapped labelled pieces cannot remain.
C_2: O00, O03, O08, O11, O22, O25, O32	$\sum_i \epsilon_i = 0 \pmod 2$	A single flipped piece cannot remain; flip errors are corrected in pairs.
C_5: O17, O28	$\sum_i t_i = 0 \pmod 5$ in each orbit separately	A single nonzero twist cannot remain. Opposite twists in different orbits do not cancel this constraint.
D_5: O06	Even total reflection bit; the abelian quotient is C_2	A single reflection cannot remain. A residual rotation can remain and is corrected with a commutator.
A_5: O34	Only the 60 attainable local orientations; no nontrivial abelian orientation-sum quotient	A single nonidentity corner orientation can remain legally and can be corrected in place.
Trivial local group	No independent sticker orientation once position is fixed	Do not prescribe a flip even when the piece has two stickers.

The O06 distinction: rotation and reflection

Write $D_5 = \langle R, F \mid R^5 = F^2 = 1,\ FRF = R^{-1}\rangle$. Here R cycles the five stickers and F reverses their cyclic order. The remaining orientation lies in $D_5' = \langle R\rangle$. Since

$$[R^k, F] = R^{2k}, \qquad 2^{-1} \equiv 3 \pmod 5,$$

every residual rotation is a commutator. To apply a needed correction R^a, choose $k \equiv 3a \pmod 5$. With this chronological convention a present residual R^b needs $a \equiv -b$.

The reflection bit is *not* the ordinary permutation sign of the five sticker labels. A pentagonal reflection swaps two pairs and has even sticker-permutation sign. Using that sign as the reflection invariant would miss the actual C_2 constraint.

The O34 distinction: a single corner twist can be legal

$A_5' = A_5$. The computed group contains 1 identity, 15 elements of order 2, 20 of order 3, and 24 of order 5. The commutator table covers all 60 elements. This does not allow arbitrary permutations of the twenty stickers; only the specified 60 local orientations are attainable. A cyclic "twist sum mod 20" is the wrong model.

5. Observed cases and the protection order

What occurred in the recorded seed-600 solve

Every orbit completed its positional permutation without an odd two-buffer residual. The final-buffer correction occurred on **O06 and O34**, using eight stars in each case. No final-buffer correction was needed on any C_2 or C_5 orbit. Twenty-seven stars corrected buffer A across nine orbits. These counts were recovered by instrumenting the same controller and requiring its entire output command sequence to equal the stored trace. [S4]

Two useful diagnostic examples follow from the independent per-orbit invariants. Swapping two pieces in O04 and two in O05 gives globally even parity, but violates each orbit's separate constraint. Similarly, flipping one O03 piece and one O08 piece does not satisfy either orbit's individual flip invariant. These are diagnostic constructions, not states observed in the verified legal scrambles.

A color-only reassignment of indistinguishable copies can fabricate an apparent odd labelled permutation. Use identities reconstructed from legal move history. An apparent parity defect can also indicate a mistaken orbit filter, wrong native move mapping, an incomplete macro, or orientation frames interpreted in the wrong order. Do not invent a parity algorithm until those possibilities have been checked.

Exact production order for the stored raw seeds

```
O06 O00 O17 O15 O02 O22 O21 O08 O23 O09
O29 O25 O13 O11 O10 O34 O28 O26 O24 O16
O14 O12 O03 O33 O32 O31 O30 O27 O20 O19
O18 O07 O05 O04 O01
```

The 12 raw seeds with no collateral are O01, O04, O05, O07, O18, O19, O20, O27, O30, O31, O32 and O33. For every other seed, its card lists *all* affected collateral orbits and the number of affected pieces and stickers. Replacing a seed or adding a different macro requires a new dependency check. Increasing rank is not a valid substitute for this order. [S3,S4]

Target seeds versus special target seeds

`target_seeds.json` is the complete 35-row working table. `target_special_seeds.json` contains four entries, for O34, O15, O17 and O29. Their stored words are **exact duplicates** of the corresponding entries in the working table. They add no extra move family and are not a set of parity algorithms. They are marked SPECIAL on the cards. The files do not provide a separate named taxonomy explaining that label; no such taxonomy is assumed here. [S3,S4]

What "three-cycle" guarantees

On its target orbit, every seed has exactly k parallel sticker three-cycles spanning three physical pieces. After three applications those target stickers return. Other orbits can have quite different cycles or in-place orientation changes. The complete effect, rather than a screenshot of a selected rank, determines purity.

6. Orbit index

Orbit	Rank	Stickers	Number	Local group	Seed length	Stage	Card	Purity
O00	1	2	1,200	C_2	2,182	2	9	collateral
O01	2	1	3,600	$\{1\}$	2,190	35	10	pure
O02	3	1	2,400	$\{1\}$	9,214	5	11	collateral
O03	3	2	3,600	C_2	1,094	23	12	collateral
O04	4	1	7,200	$\{1\}$	546	34	13	pure
O05	4	1	7,200	$\{1\}$	546	33	14	pure
O06	4	5	720	D_5	382	1	15	collateral
O07	5	1	7,200	$\{1\}$	286	32	16	pure
O08	5	2	3,600	C_2	1,534	8	17	collateral
O09	6	2	7,200	$\{1\}$	382	10	18	collateral
O10	7	1	7,200	$\{1\}$	766	15	19	collateral
O11	7	2	3,600	C_2	766	14	20	collateral
O12	8	1	7,200	$\{1\}$	766	22	21	collateral
O13	8	1	7,200	$\{1\}$	142	13	22	collateral
O14	8	2	7,200	$\{1\}$	1,534	21	23	collateral
O15*	9	1	2,400	$\{1\}$	11,564	4	24	collateral
O16	9	2	7,200	$\{1\}$	286	20	25	collateral
O17*	9	5	1,440	C_5	1,534	3	26	collateral
O18	10	1	7,200	$\{1\}$	1,534	31	27	pure
O19	10	1	7,200	$\{1\}$	142	30	28	pure
O20	10	2	7,200	$\{1\}$	142	29	29	pure
O21	11	1	7,200	$\{1\}$	94	7	30	collateral
O22	11	2	3,600	C_2	766	6	31	collateral
O23	12	2	7,200	$\{1\}$	46	9	32	collateral
O24	13	1	7,200	$\{1\}$	46	19	33	collateral
O25	13	2	3,600	C_2	46	12	34	collateral
O26	14	1	7,200	$\{1\}$	46	18	35	collateral
O27	14	1	7,200	$\{1\}$	94	28	36	pure
O28	14	5	1,440	C_5	3,070	17	37	collateral
O29*	15	1	2,400	$\{1\}$	9,214	11	38	collateral
O30	15	2	7,200	$\{1\}$	94	27	39	pure
O31	16	1	7,200	$\{1\}$	142	26	40	pure
O32	17	2	3,600	C_2	94	25	41	pure
O33	18	1	2,400	$\{1\}$	46	24	42	pure
O34*	19	20	120	A_5	12,286	16	43	collateral

$*$ denotes membership in the four-row special-seed file. The fixed centers contribute 600 additional singleton position orbits; they need no algorithms. Identical rank and sticker count do not identify the same orbit: O04/O05, O12/O13, O18/O19, and O26/O27 are distinct pairs. [S3,S4]

7. How to use an algorithm card

Read the descriptor first. Confirm the true orbit, not merely rank or sticker count. All IDs on a card refer to the retained model. The raw target table lists the three piece positions cycled by the unrelocated seed. $H(P)$ gives sticker-hosting cells; $M(P)$ is the full cap signature that distinguishes positions sharing colors or rank.

Record the exact seed once. The displayed formula is a compact definition; the text file contains the complete reduced native body and inverse. Its native primitive dictionary covers all distinct H_c, T_c appearing in the seed or relocation. For an order-three token, exchange angles 1 and 2 to invert it. Numeric move tokens are an addressing convention, not newly installed GUI keybindings.

Use the relocated controller for target selection. The card's R_o, buffer positions and third position define $K_o = R_o^{-1} W_o R_o$. Request a tree node for the desired target and orientation, then execute $S^{-1} K_o S$. A node's full ordered frame is available in `execution_trees.json.gz`; a position alone does not specify an oriented star.

Respect all collateral. A pure raw seed fixes every other orbit exactly at completion. For a nonpure seed, each support-table row counts pieces with at least one displaced sticker, including pieces twisted in place. All orbits absent from that table and the target row are fixed by the complete seed. Conjugating moves the locations of its support but keeps the same affected orbit families.

Finish orientation and audit. Use the generic position and orientation procedures on page 4. Keep the two buffers until the end of the orbit. Check the entire target orbit and all previously completed orbits after the stage. Hiding a piece never exempts it from the solved test.

> **Example: the raw O33 seed.** $W_{33} = \mathcal{C}(U; H_{15}, H_{56}, H_{21})$ expands to 46 stored moves and exactly the sticker cycle $(3029\ 38102\ 48927)$. The relocation adds ten moves, so its reference star has length 56. A target-specific setup of depth d gives $56 + 2d$ moves. Both directions count as one star. Its purity makes this a useful initial test of the addressing and macro machinery. [S3,S4]

Limits of this reference. The cards specify executable words and exact effects for the toolkit; they do not claim that the original GUI already exposes orbit IDs, frame-node selection, or atomic macro evaluation. Those features need the enhanced client described in the earlier toolchain specification. The current words are not claimed to be shortest or human-efficient.

O00 / Rank 1

TARGET SEED / STAGE 02 OF 35 / COLLATERAL REQUIRES STAGE ORDER

1,200 pieces	2 stickers per piece	Local group C_2	2,182 stored moves

$$W_{00} = \mathcal{C}(Z^8; T_{14}, T_{18}).$$

Use $U = [H_0, T_1]$, $V = [H_0, T_5]$, and $Z = [[U^4, T_{18}], H_0]$. The displayed expression has 2,246 moves before local cancellation.

Target effect $(P_{117}\ P_{4939}\ P_{6290})$, with exactly 2 sticker three-cycles. One is $(117\ 7799\ 7911)$. All other pieces in O00 are fixed.

Piece ID	Hosting cells $H(P)$	Cap signature $M(P)$
117	$\{0, 1\}$	$\{0, 1\}$
4939	$\{14, 18\}$	$\{14, 18\}$
6290	$\{18, 23\}$	$\{18, 23\}$

Relocate to the two-buffer controller

$$\begin{aligned} R_{00} = {} & T_{14}\, H_{41}\, T_{102}\, T_{114}\, H_{114}\, H_{196}\, H_{281}\, H_{342}\, H_{425} \\ & H_{488}\, H_{529}\, H_{562}\, H_{564}\, H_{557}\, T_{568}\, H_{574}\, T_{595}\, H_{595} \\ (A, B, C) = {} & (117, 175946, 6290). \end{aligned}$$

The reference star has 2,218 moves. Setup depths range from 0 to 20; all 2,396 required oriented nonbuffer states are reached. A node of depth d costs $2,218 + 2d$ moves.

Orientation and endgame Place all pieces, then correct flips in pairs with the two-star transfer. Total flip parity is even within this orbit. After the other pieces and buffer A are oriented, buffer B must be correct.

Complete collateral The full seed moves 1,272 stickers. In addition to the three target pieces, the following orbits are affected. Entries are orbit / pieces / stickers.

O01 / 6 / 6
O03 / 3 / 6
O04 / 16 / 16
O05 / 6 / 6
O07 / 10 / 10
O08 / 11 / 22
O09 / 20 / 40
O10 / 22 / 22
O11 / 13 / 26
O12 / 29 / 29
O13 / 23 / 23
O14 / 31 / 62
O15 / 8 / 8
O16 / 36 / 72
O17 / 9 / 45
O18 / 31 / 31
O19 / 42 / 42
O20 / 41 / 82
O21 / 40 / 40
O22 / 17 / 34
O23 / 39 / 78
O24 / 51 / 51
O25 / 22 / 44
O26 / 55 / 55
O27 / 53 / 53
O28 / 12 / 60
O29 / 18 / 18
O30 / 61 / 122
O31 / 61 / 61
O32 / 31 / 62
O33 / 20 / 20
O34 / 1 / 20

Native primitive dictionary Positive moves; invert order-three tokens by replacing angle 1 with 2, or 2 with 1.

H_0 = `0:0:1:1`
T_1 = `1:1:1:1`
T_{14} = `14:1:1:1`
T_{18} = `18:1:1:1`
H_{41} = `41:0:1:1`
T_{102} = `102:1:1:1`
H_{114} = `114:0:1:1`
T_{114} = `114:1:1:1`
H_{196} = `196:0:1:1`
H_{281} = `281:0:1:1`
H_{342} = `282:0:1:4`
H_{425} = `197:0:1:4`
H_{488} = `115:0:1:4`
H_{529} = `87:6:1:4`
H_{557} = `58:2:1:4`
H_{562} = `37:0:1:4`
H_{564} = `42:0:1:4`
T_{568} = `25:5:1:4`
H_{574} = `21:0:1:4`
H_{595} = `1:0:1:4`
T_{595} = `1:5:1:4`

Exact bodies: `O00_exact_words.txt`. Full frames/effects: `seed_atlas.json`. Seed word hash: `8e4a7aa2f97b42f7b051`. [S3,S4]

O01 / Rank 2

TARGET SEED / STAGE 35 OF 35 / PURE RAW EFFECT

3,600 pieces	1 sticker per piece	Local group $\{1\}$	2,190 stored moves

$$W_{01} = \mathcal{C}(Z^4; H_{27}, H_1, H_{122}).$$

Use $U = [H_0, T_1]$, $V = [H_0, T_5]$, and $Z = [[U^4, T_{18}], H_0]$. The displayed expression has 2,254 moves before local cancellation.

Target effect $(P_{536}\ P_{23235}\ P_{54890})$, with exactly 1 sticker three-cycle. One is $(563\ 31179\ 76327)$. All other pieces in O01 are fixed.

Piece ID	Hosting cells $H(P)$	Cap signature $M(P)$
536	$\{1\}$	$\{1, 3, 18\}$
23235	$\{72\}$	$\{27, 72, 122\}$
54890	$\{176\}$	$\{122, 176, 267\}$

Relocate to the two-buffer controller

$$R_{01} = T_{27}\, T_{27}\, H_{75}\, H_{189}\, T_{348}\, H_{348}\, H_{450}\, H_{534}\, H_{577}$$
$$T_{568}\, T_{597}$$
$$(A, B, C) = (536, 176045, 54890).$$

The reference star has 2,212 moves. Setup depths range from 0 to 12; all 3,598 required oriented nonbuffer states are reached. A node of depth d costs $2,212 + 2d$ moves.

Orientation and endgame There is no independent local orientation. Once a legally tracked piece is in its correct position, its sticker arrangement is determined. Use the position insertion routine; do not add a flip algorithm.

Complete support Exactly three target pieces and 3 target stickers move. Every other orbit is fixed at completion.

Native primitive dictionary Positive moves; invert order-three tokens by replacing angle 1 with 2, or 2 with 1.

H_0 = `0:0:1:1`	H_{75} = `75:0:1:1`	H_{534} = `66:0:1:4`
H_1 = `1:0:1:1`	H_{122} = `122:0:1:1`	T_{568} = `25:5:1:4`
T_1 = `1:1:1:1`	H_{189} = `189:0:1:1`	H_{577} = `16:0:1:4`
T_{18} = `18:1:1:1`	H_{348} = `270:0:1:4`	T_{597} = `3:5:1:4`
H_{27} = `27:0:1:1`	T_{348} = `270:5:1:4`	
T_{27} = `27:1:1:1`	H_{450} = `158:6:1:4`	

Exact bodies: `001_exact_words.txt`. Full frames/effects: `seed_atlas.json`. Seed word hash: `d0d775bfdfa24a27e982`. [S3,S4]

O02 / Rank 3

TARGET SEED / STAGE 05 OF 35 / COLLATERAL REQUIRES STAGE ORDER

2,400 pieces	1 sticker per piece	Local group $\{1\}$	9,214 stored moves

$$W_{02} = \mathcal{C}(U^4; T_{18}, T_{17}, H_{10}, T_{37}, H_{38}, H_{58}, H_{45}, H_{43}, T_{45}).$$

Use $U = [H_0, T_1]$, $V = [H_0, T_5]$, and $Z = [[U^4, T_{18}], H_0]$.

Target effect $(P_{12966}\ P_{14260}\ P_{29233})$, with exactly 1 sticker three-cycle. One is $(17021\ 18742\ 39537)$. All other pieces in O02 are fixed.

Piece ID	Hosting cells $H(P)$	Cap signature $M(P)$
12966	$\{39\}$	$\{8, 39, 45, 49\}$
14260	$\{43\}$	$\{27, 33, 43, 104\}$
29233	$\{91\}$	$\{45, 67, 74, 91\}$

Relocate to the two-buffer controller

$$R_{02} = T_{27}\ H_{25}\ H_{83}\ T_{204}\ H_{287}\ H_{354}\ H_{462}\ T_{540}$$
$$(A, B, C) = (12966, 163655, 29233).$$

The reference star has 9,230 moves. Setup depths range from 0 to 10; all 2,398 required oriented nonbuffer states are reached. A node of depth d costs $9,230 + 2d$ moves.

Orientation and endgame There is no independent local orientation. Once a legally tracked piece is in its correct position, its sticker arrangement is determined. Use the position insertion routine; do not add a flip algorithm.

Complete collateral The full seed moves 195 stickers. In addition to the three target pieces, the following orbits are affected. Entries are orbit / pieces / stickers.

- O04 / 3 / 3
- O05 / 6 / 6
- O08 / 3 / 6
- O18 / 3 / 3
- O19 / 6 / 6
- O20 / 3 / 6
- O21 / 9 / 9
- O22 / 6 / 12
- O23 / 12 / 24
- O24 / 15 / 15
- O25 / 6 / 12
- O26 / 12 / 12
- O27 / 12 / 12
- O28 / 3 / 15
- O29 / 3 / 3
- O30 / 11 / 22
- O31 / 8 / 8
- O32 / 6 / 12
- O33 / 6 / 6

Native primitive dictionary Positive moves; invert order-three tokens by replacing angle 1 with 2, or 2 with 1.

H_0 = `0:0:1:1`
T_1 = `1:1:1:1`
H_{10} = `10:0:1:1`
T_{17} = `17:1:1:1`
T_{18} = `18:1:1:1`
H_{25} = `25:0:1:1`
T_{27} = `27:1:1:1`
T_{37} = `37:1:1:1`
H_{38} = `38:0:1:1`
H_{43} = `43:0:1:1`
H_{45} = `45:0:1:1`
T_{45} = `45:1:1:1`
H_{58} = `58:0:1:1`
H_{83} = `83:0:1:1`
T_{204} = `204:1:1:1`
H_{287} = `286:0:1:1`
H_{354} = `268:2:1:4`
H_{462} = `178:6:1:4`
T_{540} = `45:5:1:4`

Exact bodies: `002_exact_words.txt`. Full frames/effects: `seed_atlas.json`. Seed word hash: `a19c4d5d2543e8281c64`. [S3,S4]

O03 / Rank 3

TARGET SEED / STAGE 23 OF 35 / COLLATERAL REQUIRES STAGE ORDER

3,600 pieces	2 stickers per piece	Local group C_2	1,094 stored moves

$$W_{03} = \mathcal{C}(Z^4; H_{61}, H_{27}).$$

Use $U = [H_0, T_1]$, $V = [H_0, T_5]$, and $Z = [[U^4, T_{18}], H_0]$. The displayed expression has 1,126 moves before local cancellation.

Target effect $(P_{4327}\ P_{9141}\ P_{40978})$, with exactly 2 sticker three-cycles. One is $(5214\ 31202\ 83378)$. All other pieces in O03 are fixed.

Piece ID	Hosting cells $H(P)$	Cap signature $M(P)$
4327	$\{12, 27\}$	$\{0, 12, 21, 27\}$
9141	$\{27, 72\}$	$\{27, 43, 72, 122\}$
40978	$\{130, 192\}$	$\{61, 130, 192, 200\}$

Relocate to the two-buffer controller

$$R_{03} = H_{122}\, H_{260}\, T_{302}\, H_{375}\, H_{477}\, H_{561}\, T_{597}\, T_{574}$$
$$(A, B, C) = (4327, 169445, 40978).$$

The reference star has 1,110 moves. Setup depths range from 0 to 10; all 7,196 required oriented nonbuffer states are reached. A node of depth d costs $1,110 + 2d$ moves.

Orientation and endgame Place all pieces, then correct flips in pairs with the two-star transfer. Total flip parity is even within this orbit. After the other pieces and buffer A are oriented, buffer B must be correct.

Complete collateral The full seed moves 12 stickers. In addition to the three target pieces, the following orbits are affected. Entries are orbit / pieces / stickers.

O04 / 3 / 3
O05 / 3 / 3

Native primitive dictionary Positive moves; invert order-three tokens by replacing angle 1 with 2, or 2 with 1.

H_0 = `0:0:1:1`	H_{122} = `122:0:1:1`	H_{561} = `48:0:1:4`
T_1 = `1:1:1:1`	H_{260} = `260:0:1:1`	T_{574} = `21:5:1:4`
T_{18} = `18:1:1:1`	T_{302} = `242:5:1:4`	T_{597} = `3:5:1:4`
H_{27} = `27:0:1:1`	H_{375} = `224:0:1:4`	
H_{61} = `61:0:1:1`	H_{477} = `131:6:1:4`	

Exact bodies: `O03_exact_words.txt`. Full frames/effects: `seed_atlas.json`. Seed word hash: `04e2fc406e85f0f8194e`. [S3,S4]

O04 / Rank 4

TARGET SEED / STAGE 34 OF 35 / PURE RAW EFFECT

7,200 pieces	1 sticker per piece	Local group $\{1\}$	546 stored moves

$$W_{04} = [Z^4, H_{72}].$$

Use $U = [H_0, T_1]$, $V = [H_0, T_5]$, and $Z = [[U^4, T_{18}], H_0]$. The displayed expression has 562 moves before local cancellation.

Target effect $(P_{7321}\ P_{9341}\ P_{24311})$, with exactly 1 sticker three-cycle. One is $(9337\ 11963\ 32720)$. All other pieces in O04 are fixed.

Piece ID	Hosting cells $H(P)$	Cap signature $M(P)$
7321	$\{21\}$	$\{0, 1, 21, 25, 27\}$
9341	$\{27\}$	$\{0, 12, 21, 27, 72\}$
24311	$\{75\}$	$\{35, 67, 72, 75, 137\}$

Relocate to the two-buffer controller

$$R_{04} = T_{12}\,T_{10}\,T_{14}\,H_{97}\,H_{299}\,H_{493}\,H_{584}\,H_{595}$$
$$(A, B, C) = (7321, 170575, 24311).$$

The reference star has 562 moves. Setup depths range from 0 to 8; all 7,198 required oriented nonbuffer states are reached. A node of depth d costs $562 + 2d$ moves.

Orientation and endgame There is no independent local orientation. Once a legally tracked piece is in its correct position, its sticker arrangement is determined. Use the position insertion routine; do not add a flip algorithm.

Complete support Exactly three target pieces and 3 target stickers move. Every other orbit is fixed at completion.

Native primitive dictionary Positive moves; invert order-three tokens by replacing angle 1 with 2, or 2 with 1.

H_0 = `0:0:1:1`	T_{14} = `14:1:1:1`	H_{299} = `296:0:1:1`
T_1 = `1:1:1:1`	T_{18} = `18:1:1:1`	H_{493} = `98:0:1:4`
T_{10} = `10:1:1:1`	H_{72} = `72:0:1:1`	H_{584} = `20:0:1:4`
T_{12} = `12:1:1:1`	H_{97} = `97:0:1:1`	H_{595} = `1:0:1:4`

Exact bodies: `O04_exact_words.txt`. Full frames/effects: `seed_atlas.json`. Seed word hash: `d2bed912ad63ae39a70f`. [S3,S4]

O05 / Rank 4

TARGET SEED / STAGE 33 OF 35 / PURE RAW EFFECT

7,200 pieces	1 sticker per piece	Local group $\{1\}$	546 stored moves

$$W_{05} = [Z^4, H_{43}].$$

Use $U = [H_0, T_1]$, $V = [H_0, T_5]$, and $Z = [[U^4, T_{18}], H_0]$. The displayed expression has 562 moves before local cancellation.

Target effect $(P_{7320}\ P_{9344}\ P_{18999})$, with exactly 1 sticker three-cycle. One is $(9336\ 11970\ 25256)$. All other pieces in O05 are fixed.

Piece ID	Hosting cells $H(P)$	Cap signature $M(P)$
7320	$\{21\}$	$\{0, 1, 21, 27, 46\}$
9344	$\{27\}$	$\{0, 12, 21, 27, 43\}$
18999	$\{58\}$	$\{25, 43, 58, 121, 128\}$

Relocate to the two-buffer controller

$$R_{05} = H_{12}\, H_{10}\, T_{39}\, H_{125}\, H_{231}\, H_{298}\, H_{465}\, T_{570}$$
$$(A, B, C) = (7320, 170500, 18999).$$

The reference star has 562 moves. Setup depths range from 0 to 8; all 7,198 required oriented nonbuffer states are reached. A node of depth d costs $562 + 2d$ moves.

Orientation and endgame There is no independent local orientation. Once a legally tracked piece is in its correct position, its sticker arrangement is determined. Use the position insertion routine; do not add a flip algorithm.

Complete support Exactly three target pieces and 3 target stickers move. Every other orbit is fixed at completion.

Native primitive dictionary Positive moves; invert order-three tokens by replacing angle 1 with 2, or 2 with 1.

H_0 = `0:0:1:1`	T_{18} = `18:1:1:1`	H_{231} = `231:0:1:1`
T_1 = `1:1:1:1`	T_{39} = `39:1:1:1`	H_{298} = `295:0:1:1`
H_{10} = `10:0:1:1`	H_{43} = `43:0:1:1`	H_{465} = `122:0:1:4`
H_{12} = `12:0:1:1`	H_{125} = `125:0:1:1`	T_{570} = `27:3:1:4`

Exact bodies: `005_exact_words.txt`. Full frames/effects: `seed_atlas.json`. Seed word hash: `b64815f88c8b32aaed44`. [S3,S4]

O06 / Rank 4

TARGET SEED / STAGE 01 OF 35 / COLLATERAL REQUIRES STAGE ORDER

720 pieces	5 stickers per piece	Local group D_5	382 stored moves

$$W_{06} = \mathcal{C}(T_0; T_{17}, T_{38}, T_{33}, H_{27}, T_{46}, T_{61}^{-1}, T_{54}).$$

Use $U = [H_0, T_1]$, $V = [H_0, T_5]$, and $Z = [[U^4, T_{18}], H_0]$.

Target effect $(P_{4944}\ P_{17705}\ P_{15155})$, with exactly 5 sticker three-cycles. One is $(6184\ 26546\ 20051)$. All other pieces in O06 are fixed.

Piece ID	Hosting cells $H(P)$	Cap signature $M(P)$
4944	$\{14, 18, 51, 54, 61\}$	$\{14, 18, 51, 54, 61\}$
17705	$\{54, 61, 65, 92, 108\}$	$\{54, 61, 65, 92, 108\}$
15155	$\{46, 51, 61, 84, 108\}$	$\{46, 51, 61, 84, 108\}$

Relocate to the two-buffer controller

$$R_{06} = H_{108}\, T_{157}\, H_{267}\, H_{384}\, H_{491}\, T_{573}\, H_{569}\, T_{531}$$
$$(A, B, C) = (4944, 153202, 29398).$$

The reference star has 398 moves. Setup depths range from 0 to 9; all 7,180 required oriented nonbuffer states are reached. A node of depth d costs $398 + 2d$ moves.

Orientation and endgame Use all ten local frames, including reflection-type orientations. Correct target orientations with transfers. A final rotation at B is legal and can need the eight-star commutator; a single final reflection is forbidden.

Complete collateral The full seed moves 1,134 stickers. In addition to the three target pieces, the following orbits are affected. Entries are orbit / pieces / stickers.

O07 / 26 / 26	O17 / 5 / 25	O26 / 49 / 49
O09 / 22 / 44	O18 / 30 / 30	O27 / 50 / 50
O10 / 21 / 21	O19 / 30 / 30	O28 / 10 / 50
O11 / 16 / 32	O20 / 36 / 72	O29 / 20 / 20
O12 / 31 / 31	O21 / 33 / 33	O30 / 52 / 104
O13 / 31 / 31	O22 / 17 / 34	O31 / 58 / 58
O14 / 23 / 46	O23 / 37 / 74	O32 / 30 / 60
O15 / 7 / 7	O24 / 45 / 45	O33 / 21 / 21
O16 / 33 / 66	O25 / 20 / 40	O34 / 1 / 20

Native primitive dictionary Positive moves; invert order-three tokens by replacing angle 1 with 2, or 2 with 1.

T_0 = `0:1:1:1`	T_{54} = `54:1:1:1`	H_{491} = `77:0:1:4`
T_{17} = `17:1:1:1`	T_{61} = `61:1:1:1`	T_{531} = `54:5:1:4`
H_{27} = `27:0:1:1`	H_{108} = `108:0:1:1`	H_{569} = `41:2:1:4`
T_{33} = `33:1:1:1`	T_{157} = `157:1:1:1`	T_{573} = `29:5:1:4`
T_{38} = `38:1:1:1`	H_{267} = `267:0:1:1`	
T_{46} = `46:1:1:1`	H_{384} = `230:6:1:4`	

Exact bodies: `006_exact_words.txt`. Full frames/effects: `seed_atlas.json`. Seed word hash: `89072dc581c88f474ca1`. [S3,S4]

O07 / Rank 5

TARGET SEED / STAGE 32 OF 35 / PURE RAW EFFECT

7,200 pieces	1 sticker per piece	Local group $\{1\}$	286 stored moves

$$W_{07} = \mathcal{C}(U^4; T_{18}, T_{17}, H_{15}, H_{37}).$$

Use $U = [H_0, T_1]$, $V = [H_0, T_5]$, and $Z = [[U^4, T_{18}], H_0]$.

Target effect $(P_{3388}\ P_{5436}\ P_{11190})$, with exactly 1 sticker three-cycle. One is $(3926\ 6754\ 14548)$. All other pieces in O07 are fixed.

Piece ID	Hosting cells $H(P)$	Cap signature $M(P)$
3388	$\{9\}$	$\{6, 9, 26, 37, 40, 50\}$
5436	$\{15\}$	$\{1, 3, 15, 18, 19, 23\}$
11190	$\{33\}$	$\{12, 17, 27, 33, 37, 43\}$

Relocate to the two-buffer controller

$$R_{07} = T_3\, T_{20}\, H_{121}\, H_{287}\, T_{444}\, H_{550}$$
$$(A, B, C) = (3388, 173556, 11190).$$

The reference star has 298 moves. Setup depths range from 0 to 8; all 7,198 required oriented nonbuffer states are reached. A node of depth d costs $298 + 2d$ moves.

Orientation and endgame There is no independent local orientation. Once a legally tracked piece is in its correct position, its sticker arrangement is determined. Use the position insertion routine; do not add a flip algorithm.

Complete support Exactly three target pieces and 3 target stickers move. Every other orbit is fixed at completion.

Native primitive dictionary Positive moves; invert order-three tokens by replacing angle 1 with 2, or 2 with 1.

H_0 = `0:0:1:1`	T_{17} = `17:1:1:1`	H_{121} = `121:0:1:1`
T_1 = `1:1:1:1`	T_{18} = `18:1:1:1`	H_{287} = `286:0:1:1`
T_3 = `3:1:1:1`	T_{20} = `20:1:1:1`	T_{444} = `145:5:1:4`
H_{15} = `15:0:1:1`	H_{37} = `37:0:1:1`	H_{550} = `50:0:1:4`

Exact bodies: `007_exact_words.txt`. Full frames/effects: `seed_atlas.json`. Seed word hash: `3b8f842ef80822c77d47`. [S3,S4]

O08 / Rank 5

TARGET SEED / STAGE 08 OF 35 / COLLATERAL REQUIRES STAGE ORDER

3,600 pieces	2 stickers per piece	Local group C_2	1,534 stored moves

$$W_{08} = \mathcal{C}(V; T_{31}, H_{17}, H_{36}, H_{60}, H_{129}, T_{232}, H_{141}, T_{244}).$$

Use $U = [H_0, T_1]$, $V = [H_0, T_5]$, and $Z = [[U^4, T_{18}], H_0]$.

Target effect $(P_{33424}\ P_{75367}\ P_{71788})$, with exactly 2 sticker three-cycles. One is $(45481\ 105892\ 105932)$. All other pieces in O08 are fixed.

Piece ID	Hosting cells $H(P)$	Cap signature $M(P)$
33424	$\{105, 141\}$	$\{44, 105, 117, 129, 141, 220\}$
75367	$\{244, 277\}$	$\{166, 188, 232, 244, 256, 277\}$
71788	$\{232, 244\}$	$\{136, 166, 232, 237, 244, 277\}$

Relocate to the two-buffer controller

$$R_{08} = H_{166}\, T_{268}\, H_{320}\, T_{493}\, T_{456}$$
$$(A, B, C) = (33424, 125632, 48364).$$

The reference star has 1,544 moves. Setup depths range from 0 to 7; all 7,196 required oriented nonbuffer states are reached. A node of depth d costs $1,544 + 2d$ moves.

Orientation and endgame Place all pieces, then correct flips in pairs with the two-star transfer. Total flip parity is even within this orbit. After the other pieces and buffer A are oriented, buffer B must be correct.

Complete collateral The full seed moves 108 stickers. In addition to the three target pieces, the following orbits are affected. Entries are orbit / pieces / stickers.

- O09 / 6 / 12
- O10 / 6 / 6
- O11 / 3 / 6
- O12 / 6 / 6
- O13 / 3 / 3
- O14 / 6 / 12
- O16 / 3 / 6
- O18 / 3 / 3
- O25 / 3 / 6
- O26 / 3 / 3
- O27 / 6 / 6
- O30 / 6 / 12
- O31 / 6 / 6
- O32 / 6 / 12
- O33 / 3 / 3

Native primitive dictionary Positive moves; invert order-three tokens by replacing angle 1 with 2, or 2 with 1.

H_0 = `0:0:1:1`	H_{60} = `60:0:1:1`	T_{244} = `244:1:1:1`
T_5 = `5:1:1:1`	H_{129} = `129:0:1:1`	T_{268} = `268:1:1:1`
H_{17} = `17:0:1:1`	H_{141} = `141:0:1:1`	H_{320} = `248:0:1:4`
T_{31} = `31:1:1:1`	H_{166} = `166:0:1:1`	T_{456} = `117:5:1:4`
H_{36} = `36:0:1:1`	T_{232} = `232:1:1:1`	T_{493} = `98:5:1:4`

Exact bodies: `008_exact_words.txt`. Full frames/effects: `seed_atlas.json`. Seed word hash: `a6da737cff38321e7e33`. [S3,S4]

O09 / Rank 6

TARGET SEED / STAGE 10 OF 35 / COLLATERAL REQUIRES STAGE ORDER

7,200 pieces	2 stickers per piece	Local group $\{1\}$	382 stored moves

$$W_{09} = \mathcal{C}(V; T_{31}, H_{17}, H_{36}, H_5, H_{60}, H_{13}).$$

Use $U = [H_0, T_1]$, $V = [H_0, T_5]$, and $Z = [[U^4, T_{18}], H_0]$.

Target effect $(P_{260}\ P_{2572}\ P_{35799})$, with exactly 2 sticker three-cycles. One is $(260\ 2886\ 48950)$. All other pieces in O09 are fixed.

Piece ID	Hosting cells $H(P)$	Cap signature $M(P)$
260	$\{0, 10\}$	$\{0, 1, 5, 8, 10, 13, 14\}$
2572	$\{6, 9\}$	$\{6, 7, 9, 11, 13, 26, 40\}$
35799	$\{113, 136\}$	$\{60, 99, 113, 136, 166, 232, 244\}$

Relocate to the two-buffer controller

$$R_{09} = H_{40}\, H_{123}\, T_{236}\, T_{300}\, H_{489}\, H_{576}$$
$$(A, B, C) = (260, 174806, 35799).$$

The reference star has 394 moves. Setup depths range from 0 to 7; all 7,198 required oriented nonbuffer states are reached. A node of depth d costs $394 + 2d$ moves.

Orientation and endgame There is no independent local orientation. Once a legally tracked piece is in its correct position, its sticker arrangement is determined. Use the position insertion routine; do not add a flip algorithm.

Complete collateral The full seed moves 39 stickers. In addition to the three target pieces, the following orbits are affected. Entries are orbit / pieces / stickers.

O10 / 3 / 3
O11 / 3 / 6
O12 / 6 / 6
O14 / 3 / 6
O31 / 3 / 3
O32 / 3 / 6
O33 / 3 / 3

Native primitive dictionary Positive moves; invert order-three tokens by replacing angle 1 with 2, or 2 with 1.

H_0 = `0:0:1:1`
H_5 = `5:0:1:1`
T_5 = `5:1:1:1`
H_{13} = `13:0:1:1`
H_{17} = `17:0:1:1`
T_{31} = `31:1:1:1`
H_{36} = `36:0:1:1`
H_{40} = `40:0:1:1`
H_{60} = `60:0:1:1`
H_{123} = `123:0:1:1`
T_{236} = `236:1:1:1`
T_{300} = `296:3:1:4`
H_{489} = `114:0:1:4`
H_{576} = `14:0:1:4`

Exact bodies: `009_exact_words.txt`. Full frames/effects: `seed_atlas.json`. Seed word hash: `d1cc041c064a2864778c`. [S3,S4]

O10 / Rank 7

TARGET SEED / STAGE 15 OF 35 / COLLATERAL REQUIRES STAGE ORDER

7,200 pieces	1 sticker per piece	Local group $\{1\}$	766 stored moves

$$W_{10} = [W_{09}, H_{28}].$$

Use $U = [H_0, T_1]$, $V = [H_0, T_5]$, and $Z = [[U^4, T_{18}], H_0]$.

Target effect $(P_{3627}\ P_{27387}\ P_{3901})$, with exactly 1 sticker three-cycle. One is $(4200\ 37056\ 4594)$. All other pieces in O10 are fixed.

Piece ID	Hosting cells $H(P)$	Cap signature $M(P)$
3627	$\{9\}$	$\{6, 7, 9, 11, 13, 26, 28, 40\}$
27387	$\{85\}$	$\{28, 47, 66, 85, 123, 158, 163, 177\}$
3901	$\{10\}$	$\{0, 1, 5, 8, 10, 13, 14, 32\}$

Relocate to the two-buffer controller

$$R_{10} = H_{123}\, H_{214}\, H_{402}\, H_{565}\, H_{599}$$
$$(A, B, C) = (3627, 173386, 3901).$$

The reference star has 776 moves. Setup depths range from 0 to 7; all 7,198 required oriented nonbuffer states are reached. A node of depth d costs $776 + 2d$ moves.

Orientation and endgame There is no independent local orientation. Once a legally tracked piece is in its correct position, its sticker arrangement is determined. Use the position insertion routine; do not add a flip algorithm.

Complete collateral The full seed moves 24 stickers. In addition to the three target pieces, the following orbits are affected. Entries are orbit / pieces / stickers.

O12 / 3 / 3
O14 / 3 / 6
O31 / 3 / 3
O32 / 3 / 6
O33 / 3 / 3

Native primitive dictionary Positive moves; invert order-three tokens by replacing angle 1 with 2, or 2 with 1.

H_0 = `0:0:1:1`	H_{28} = `28:0:1:1`	H_{214} = `214:0:1:1`
H_5 = `5:0:1:1`	T_{31} = `31:1:1:1`	H_{402} = `195:0:1:4`
T_5 = `5:1:1:1`	H_{36} = `36:0:1:1`	H_{565} = `38:6:1:4`
H_{13} = `13:0:1:1`	H_{60} = `60:0:1:1`	H_{599} = `13:2:1:4`
H_{17} = `17:0:1:1`	H_{123} = `123:0:1:1`	

Exact bodies: `O10_exact_words.txt`. Full frames/effects: `seed_atlas.json`. Seed word hash: `71ec0bde706399db56e7`. [S3,S4]

O11 / Rank 7

TARGET SEED / STAGE 14 OF 35 / COLLATERAL REQUIRES STAGE ORDER

3,600 pieces	2 stickers per piece	Local group C_2	766 stored moves

$$W_{11} = [W_{09}, H_{50}].$$

Use $U = [H_0, T_1]$, $V = [H_0, T_5]$, and $Z = [[U^4, T_{18}], H_0]$.

Target effect $(P_{261}\ P_{2573}\ P_{48143})$, with exactly 2 sticker three-cycles. One is $(261\ 2887\ 71319)$. All other pieces in O11 are fixed.

Piece ID	Hosting cells $H(P)$	Cap signature $M(P)$
261	$\{0, 10\}$	$\{0, 1, 5, 8, 10, 13, 14, 18\}$
2573	$\{6, 9\}$	$\{6, 7, 9, 11, 13, 26, 40, 50\}$
48143	$\{153, 164\}$	$\{50, 73, 107, 153, 164, 165, 229, 245\}$

Relocate to the two-buffer controller

$$R_{11} = H_{40}\, T_{123}\, H_{332}\, T_{492}\, H_{576}$$
$$(A, B, C) = (261, 174807, 48143).$$

The reference star has 776 moves. Setup depths range from 0 to 7; all 7,196 required oriented nonbuffer states are reached. A node of depth d costs $776 + 2d$ moves.

Orientation and endgame Place all pieces, then correct flips in pairs with the two-star transfer. Total flip parity is even within this orbit. After the other pieces and buffer A are oriented, buffer B must be correct.

Complete collateral The full seed moves 24 stickers. In addition to the three target pieces, the following orbits are affected. Entries are orbit / pieces / stickers.

O12 / 6 / 6
O31 / 3 / 3
O32 / 3 / 6
O33 / 3 / 3

Native primitive dictionary Positive moves; invert order-three tokens by replacing angle 1 with 2, or 2 with 1.

H_0 = `0:0:1:1`
H_5 = `5:0:1:1`
T_5 = `5:1:1:1`
H_{13} = `13:0:1:1`
H_{17} = `17:0:1:1`
T_{31} = `31:1:1:1`
H_{36} = `36:0:1:1`
H_{40} = `40:0:1:1`
H_{50} = `50:0:1:1`
H_{60} = `60:0:1:1`
T_{123} = `123:1:1:1`
H_{332} = `254:0:1:4`
T_{492} = `97:5:1:4`
H_{576} = `14:0:1:4`

Exact bodies: `011_exact_words.txt`. Full frames/effects: `seed_atlas.json`. Seed word hash: `befd7adc70177ca5630c`. [S3,S4]

O12 / Rank 8

TARGET SEED / STAGE 22 OF 35 / COLLATERAL REQUIRES STAGE ORDER

7,200 pieces	1 sticker per piece	Local group $\{1\}$	766 stored moves

$$W_{12} = [W_{09}, H_{74}].$$

Use $U = [H_0, T_1]$, $V = [H_0, T_5]$, and $Z = [[U^4, T_{18}], H_0]$.

Target effect $(P_{2574}\ P_{35825}\ P_{52109})$, with exactly 1 sticker three-cycle. One is $(2888\ 48985\ 72244)$. All other pieces in O12 are fixed.

Piece ID	Hosting cells $H(P)$	Cap signature $M(P)$
2574	$\{6\}$	$\{6, 7, 9, 11, 13, 26, 30, 40, 50\}$
35825	$\{113\}$	$\{56, 60, 74, 99, 113, 136, 166, 232, 244\}$
52109	$\{166\}$	$\{74, 107, 113, 154, 164, 166, 171, 198, 212\}$

Relocate to the two-buffer controller

$$R_{12} = H_{99}\, H_{183}\, H_{375}\, H_{553}\, T_{589}$$
$$(A, B, C) = (2574, 175510, 52109).$$

The reference star has 776 moves. Setup depths range from 0 to 6; all 7,198 required oriented nonbuffer states are reached. A node of depth d costs $776 + 2d$ moves.

Orientation and endgame There is no independent local orientation. Once a legally tracked piece is in its correct position, its sticker arrangement is determined. Use the position insertion routine; do not add a flip algorithm.

Complete collateral The full seed moves 6 stickers. In addition to the three target pieces, the following orbits are affected. Entries are orbit / pieces / stickers.

O33 / 3 / 3

Native primitive dictionary Positive moves; invert order-three tokens by replacing angle 1 with 2, or 2 with 1.

H_0 = `0:0:1:1`
H_5 = `5:0:1:1`
T_5 = `5:1:1:1`
H_{13} = `13:0:1:1`
H_{17} = `17:0:1:1`
T_{31} = `31:1:1:1`
H_{36} = `36:0:1:1`
H_{60} = `60:0:1:1`
H_{74} = `74:0:1:1`
H_{99} = `99:0:1:1`
H_{183} = `183:0:1:1`
H_{375} = `224:0:1:4`
H_{553} = `52:6:1:4`
T_{589} = `11:5:1:4`

Exact bodies: `O12_exact_words.txt`. Full frames/effects: `seed_atlas.json`. Seed word hash: `199a3bbe2fc0851db74b`. [S3,S4]

O13 / Rank 8

TARGET SEED / STAGE 13 OF 35 / COLLATERAL REQUIRES STAGE ORDER

7,200 pieces	1 sticker per piece	Local group $\{1\}$	142 stored moves

$$W_{13} = \mathcal{C}(U^4; T_{18}, T_{17}, H_{127}).$$

Use $U = [H_0, T_1]$, $V = [H_0, T_5]$, and $Z = [[U^4, T_{18}], H_0]$.

Target effect $(P_{9394}\ P_{18846}\ P_{37809})$, with exactly 1 sticker three-cycle. One is $(12066\ 24980\ 51831)$. All other pieces in O13 are fixed.

Piece ID	Hosting cells $H(P)$	Cap signature $M(P)$
9394	$\{27\}$	$\{0, 1, 12, 17, 21, 25, 27, 43, 58\}$
18846	$\{57\}$	$\{18, 23, 41, 57, 69, 102, 119, 127, 138\}$
37809	$\{119\}$	$\{23, 57, 69, 76, 119, 120, 127, 167, 174\}$

Relocate to the two-buffer controller

$$R_{13} = H_{41}\, H_{79}\, H_{271}\, H_{467}\, H_{554}$$
$$(A, B, C) = (9394, 169498, 37809).$$

The reference star has 152 moves. Setup depths range from 0 to 6; all 7,198 required oriented nonbuffer states are reached. A node of depth d costs $152 + 2d$ moves.

Orientation and endgame There is no independent local orientation. Once a legally tracked piece is in its correct position, its sticker arrangement is determined. Use the position insertion routine; do not add a flip algorithm.

Complete collateral The full seed moves 15 stickers. In addition to the three target pieces, the following orbits are affected. Entries are orbit / pieces / stickers.

O16 / 3 / 6
O20 / 3 / 6

Native primitive dictionary Positive moves; invert order-three tokens by replacing angle 1 with 2, or 2 with 1.

H_0 = `0:0:1:1`
T_1 = `1:1:1:1`
T_{17} = `17:1:1:1`
T_{18} = `18:1:1:1`
H_{41} = `41:0:1:1`
H_{79} = `79:0:1:1`
H_{127} = `127:0:1:1`
H_{271} = `271:0:1:1`
H_{467} = `121:0:1:4`
H_{554} = `43:0:1:4`

Exact bodies: `O13_exact_words.txt`. Full frames/effects: `seed_atlas.json`. Seed word hash: `38b53a036bc5fb44e687`. [S3,S4]

O14 / Rank 8

TARGET SEED / STAGE 21 OF 35 / COLLATERAL REQUIRES STAGE ORDER

7,200 pieces	2 stickers per piece	Local group $\{1\}$	1,534 stored moves

$$W_{14} = \mathcal{C}(V; T_{31}, H_{17}, H_{36}, H_{60}, H_{129}, T_{232}, H_{146}, H_{90}).$$

Use $U = [H_0, T_1]$, $V = [H_0, T_5]$, and $Z = [[U^4, T_{18}], H_0]$.

Target effect $(P_{24913}\ P_{57802}\ P_{40862})$, with exactly 2 sticker three-cycles. One is $(33606\ 91107\ 61337)$. All other pieces in O14 are fixed.

Piece ID	Hosting cells $H(P)$	Cap signature $M(P)$
24913	$\{77, 134\}$	$\{29, 64, 68, 70, 77, 90, 101, 111, 134\}$
57802	$\{186, 210\}$	$\{129, 141, 146, 186, 199, 210, 220, 293, 318\}$
40862	$\{129, 141\}$	$\{59, 90, 105, 129, 141, 146, 177, 186, 210\}$

Relocate to the two-buffer controller

$$R_{14} = T_{141}\, T_{117}\, H_{180}\, H_{420}\, H_{491}$$
$$(A, B, C) = (24913, 143454, 44557).$$

The reference star has 1,544 moves. Setup depths range from 0 to 6; all 7,198 required oriented nonbuffer states are reached. A node of depth d costs $1,544 + 2d$ moves.

Orientation and endgame There is no independent local orientation. Once a legally tracked piece is in its correct position, its sticker arrangement is determined. Use the position insertion routine; do not add a flip algorithm.

Complete collateral The full seed moves 15 stickers. In addition to the three target pieces, the following orbits are affected. Entries are orbit / pieces / stickers.

O32 / 3 / 6
O33 / 3 / 3

Native primitive dictionary Positive moves; invert order-three tokens by replacing angle 1 with 2, or 2 with 1.

H_0 = `0:0:1:1`	H_{60} = `60:0:1:1`	H_{146} = `146:0:1:1`
T_5 = `5:1:1:1`	H_{90} = `90:0:1:1`	H_{180} = `180:0:1:1`
H_{17} = `17:0:1:1`	T_{117} = `117:1:1:1`	T_{232} = `232:1:1:1`
T_{31} = `31:1:1:1`	H_{129} = `129:0:1:1`	H_{420} = `155:0:1:4`
H_{36} = `36:0:1:1`	T_{141} = `141:1:1:1`	H_{491} = `77:0:1:4`

Exact bodies: `014_exact_words.txt`. Full frames/effects: `seed_atlas.json`. Seed word hash: `1a43982e6af1ff4f004b`. [S3,S4]

O15 / Rank 9

SPECIAL TARGET SEED / STAGE 04 OF 35 / COLLATERAL REQUIRES STAGE ORDER

2,400 pieces	**1 sticker per piece**	**Local group $\{1\}$**	**11,564 stored moves**

$$W_{15} = \mathcal{C}(T_0;T_{19}, T_{57}, T_0, T_{24}, T_0, T_{24}, T_0^{-1}, T_{24}, T_0^{-1}, T_{24}, T_0, T_{24}).$$

Use $U = [H_0, T_1]$, $V = [H_0, T_5]$, and $Z = [[U^4, T_{18}], H_0]$. The displayed expression has 12,286 moves before local cancellation.

Target effect $(P_{1899}\ P_{5529}\ P_{3322})$, with exactly 1 sticker three-cycle. One is $(2109\ 6872\ 3841)$. All other pieces in O15 are fixed.

Piece ID	Hosting cells $H(P)$	Cap signature $M(P)$
1899	$\{4\}$	$\{2, 4, 5, 7, 8, 13, 19, 24, 29, 32\}$
5529	$\{15\}$	$\{1, 2, 3, 7, 15, 18, 19, 23, 24, 57\}$
3322	$\{8\}$	$\{0, 4, 5, 8, 10, 13, 14, 29, 32, 41\}$

Relocate to the two-buffer controller

$$R_{15} = H_{23}\, H_{147}\, H_{370}\, H_{550}\, T_{594}$$
$$(A, B, C) = (1899, 175095, 3322).$$

The reference star has 11,574 moves. Setup depths range from 0 to 6; all 2,398 required oriented nonbuffer states are reached. A node of depth d costs $11,574 + 2d$ moves.

Orientation and endgame There is no independent local orientation. Once a legally tracked piece is in its correct position, its sticker arrangement is determined. Use the position insertion routine; do not add a flip algorithm.

Complete collateral The full seed moves 647 stickers. In addition to the three target pieces, the following orbits are affected. Entries are orbit / pieces / stickers.

- O10 / 3 / 3
- O11 / 3 / 6
- O12 / 6 / 6
- O13 / 9 / 9
- O14 / 3 / 6
- O16 / 9 / 18
- O18 / 23 / 23
- O19 / 18 / 18
- O21 / 8 / 8
- O22 / 17 / 34
- O23 / 24 / 48
- O24 / 32 / 32
- O25 / 15 / 30
- O26 / 41 / 41
- O27 / 46 / 46
- O28 / 10 / 50
- O29 / 16 / 16
- O30 / 48 / 96
- O31 / 56 / 56
- O32 / 29 / 58
- O33 / 20 / 20
- O34 / 1 / 20

Native primitive dictionary Positive moves; invert order-three tokens by replacing angle 1 with 2, or 2 with 1.

- T_0 = `0:1:1:1`
- T_{19} = `19:1:1:1`
- H_{23} = `23:0:1:1`
- T_{24} = `24:1:1:1`
- T_{57} = `57:1:1:1`
- H_{147} = `147:0:1:1`
- H_{370} = `229:0:1:4`
- H_{550} = `50:0:1:4`
- T_{594} = `5:5:1:4`

Exact bodies: `015_exact_words.txt`. Full frames/effects: `seed_atlas.json`. Seed word hash: `eae678a948d60e2dfaf5`. [S3,S4]

O16 / Rank 9

TARGET SEED / STAGE 20 OF 35 / COLLATERAL REQUIRES STAGE ORDER

7,200 pieces	2 stickers per piece	Local group $\{1\}$	286 stored moves

$$W_{16} = \mathcal{C}(V^4; T_{14}, H_{38}, H_{24}, H_{167}).$$

Use $U = [H_0, T_1]$, $V = [H_0, T_5]$, and $Z = [[U^4, T_{18}], H_0]$.

Target effect $(P_{3235}\ P_{26508}\ P_{54119})$, with exactly 2 sticker three-cycles. One is $(3731\ 64381\ 75090)$. All other pieces in O16 are fixed.

Piece ID	Hosting cells $H(P)$	Cap signature $M(P)$
3235	$\{8, 32\}$	$\{4, 5, 8, 10, 13, 14, 24, 29, 32, 41\}$
26508	$\{82, 148\}$	$\{24, 52, 62, 70, 82, 120, 131, 148, 167, 174\}$
54119	$\{173, 184\}$	$\{119, 127, 138, 167, 173, 184, 211, 264, 291, 328\}$

Relocate to the two-buffer controller

$$R_{16} = H_{52}\, H_{252}\, H_{474}\, T_{560}\, T_{590}$$
$$(A, B, C) = (3235, 170747, 54119).$$

The reference star has 296 moves. Setup depths range from 0 to 6; all 7,198 required oriented nonbuffer states are reached. A node of depth d costs $296 + 2d$ moves.

Orientation and endgame There is no independent local orientation. Once a legally tracked piece is in its correct position, its sticker arrangement is determined. Use the position insertion routine; do not add a flip algorithm.

Complete collateral The full seed moves 9 stickers. In addition to the three target pieces, the following orbits are affected. Entries are orbit / pieces / stickers.

O19 / 3 / 3

Native primitive dictionary Positive moves; invert order-three tokens by replacing angle 1 with 2, or 2 with 1.

H_0 = `0:0:1:1`	H_{38} = `38:0:1:1`	H_{474} = `133:6:1:4`
T_5 = `5:1:1:1`	H_{52} = `52:0:1:1`	T_{560} = `35:5:1:4`
T_{14} = `14:1:1:1`	H_{167} = `167:0:1:1`	T_{590} = `10:5:1:4`
H_{24} = `24:0:1:1`	H_{252} = `252:0:1:1`	

Exact bodies: `016_exact_words.txt`. Full frames/effects: `seed_atlas.json`. Seed word hash: `1a00a2ce4064d6fa2b89`. [S3,S4]

O17 / Rank 9

SPECIAL TARGET SEED / STAGE 03 OF 35 / COLLATERAL REQUIRES STAGE ORDER

1,440 pieces	5 stickers per piece	Local group C_5	1,534 stored moves

$$W_{17} = \mathcal{C}(T_0;T_{19}, T_{10}, T_2, T_{14}, T_4,\\ T_{18}, T_4, T_{18}, T_4).$$

Use $U = [H_0, T_1]$, $V = [H_0, T_5]$, and $Z = [[U^4, T_{18}], H_0]$.

Target effect $(P_{269}\ P_{1062}\ P_{398})$, with exactly 5 sticker three-cycles. One is $(269\ 6893\ 5952)$. All other pieces in O17 are fixed.

Piece ID	Hosting cells $H(P)$	Cap signature $M(P)$
269	$\{0, 5, 8, 10, 13\}$	$\{0, 1, 4, 5, 7, 8, 10, 13, 14, 32\}$
1062	$\{2, 3, 7, 15, 19\}$	$\{1, 2, 3, 4, 7, 13, 15, 19, 23, 24\}$
398	$\{0, 1, 3, 7, 13\}$	$\{0, 1, 2, 3, 5, 7, 10, 13, 15, 18\}$

Relocate to the two-buffer controller

$$R_{17} = H_2\, H_{11}\, T_{37}\, H_{197}\, H_{426}\, H_{575}$$
$$(A, B, C) = (269, 173788, 5451).$$

The reference star has 1,546 moves. Setup depths range from 0 to 6; all 7,190 required oriented nonbuffer states are reached. A node of depth d costs $1,546 + 2d$ moves.

Orientation and endgame Place all pieces, then transfer opposite cyclic twists between a target and buffer B. Use the transported five-way frames. The total twist is zero modulo 5 within this orbit; the final buffer needs no residual correction.

Complete collateral The full seed moves 720 stickers. In addition to the three target pieces, the following orbits are affected. Entries are orbit / pieces / stickers.

- O14 / 3 / 6
- O16 / 3 / 6
- O18 / 13 / 13
- O19 / 18 / 18
- O20 / 21 / 42
- O21 / 30 / 30
- O22 / 12 / 24
- O23 / 35 / 70
- O24 / 38 / 38
- O25 / 23 / 46
- O26 / 48 / 48
- O27 / 50 / 50
- O28 / 9 / 45
- O29 / 17 / 17
- O30 / 49 / 98
- O31 / 56 / 56
- O32 / 29 / 58
- O33 / 20 / 20
- O34 / 1 / 20

Native primitive dictionary Positive moves; invert order-three tokens by replacing angle 1 with 2, or 2 with 1.

- T_0 = `0:1:1:1`
- H_2 = `2:0:1:1`
- T_2 = `2:1:1:1`
- T_4 = `4:1:1:1`
- T_{10} = `10:1:1:1`
- H_{11} = `11:0:1:1`
- T_{14} = `14:1:1:1`
- T_{18} = `18:1:1:1`
- T_{19} = `19:1:1:1`
- T_{37} = `37:1:1:1`
- H_{197} = `197:0:1:1`
- H_{426} = `196:0:1:4`
- H_{575} = `32:0:1:4`

Exact bodies: `017_exact_words.txt`. Full frames/effects: `seed_atlas.json`. Seed word hash: `b429aabf1512e024639c`. [S3,S4]

O18 / Rank 10

TARGET SEED / STAGE 31 OF 35 / PURE RAW EFFECT

7,200 pieces	**1 sticker per piece**	**Local group $\{1\}$**	**1,534 stored moves**

$$W_{18} = \mathcal{C}(V; T_{31}, H_{17}, H_{36}, H_{60}, H_{50}, H_{146}, H_{129}, T_{154}).$$

Use $U = [H_0, T_1]$, $V = [H_0, T_5]$, and $Z = [[U^4, T_{18}], H_0]$.

Target effect $(P_{36006}\ P_{61749}\ P_{68044})$, with exactly 1 sticker three-cycle. One is $(49241\ 86125\ 95346)$. All other pieces in O18 are fixed.

Piece ID	Hosting cells $H(P)$	Cap signature $M(P)$
36006	$\{113\}$	$\{26, 50, 60, 74, 107, 113, 154, 164, 166, 171, 212\}$
61749	$\{198\}$	$\{74, 89, 100, 145, 153, 154, 164, 166, 171, 198, 212\}$
68044	$\{220\}$	$\{129, 141, 165, 186, 210, 220, 245, 312, 318, 341, 365\}$

Relocate to the two-buffer controller

$$R_{18} = H_{89}\ H_{207}\ H_{326}\ T_{398}\ T_{496}\ H_{550}$$
$$(A, B, C) = (36006, 148492, 68044).$$

The reference star has 1,546 moves. Setup depths range from 0 to 6; all 7,198 required oriented nonbuffer states are reached. A node of depth d costs $1,546 + 2d$ moves.

Orientation and endgame There is no independent local orientation. Once a legally tracked piece is in its correct position, its sticker arrangement is determined. Use the position insertion routine; do not add a flip algorithm.

Complete support Exactly three target pieces and 3 target stickers move. Every other orbit is fixed at completion.

Native primitive dictionary Positive moves; invert order-three tokens by replacing angle 1 with 2, or 2 with 1.

H_0 = `0:0:1:1`
T_5 = `5:1:1:1`
H_{17} = `17:0:1:1`
T_{31} = `31:1:1:1`
H_{36} = `36:0:1:1`
H_{50} = `50:0:1:1`
H_{60} = `60:0:1:1`
H_{89} = `89:0:1:1`
H_{129} = `129:0:1:1`
H_{146} = `146:0:1:1`
T_{154} = `154:1:1:1`
H_{207} = `207:0:1:1`
H_{326} = `291:0:1:4`
T_{398} = `226:3:1:4`
T_{496} = `88:5:1:4`
H_{550} = `50:0:1:4`

Exact bodies: `018_exact_words.txt`. Full frames/effects: `seed_atlas.json`. Seed word hash: `22b13b1dc617cc7f05c7`. [S3,S4]

O19 / Rank 10

TARGET SEED / STAGE 30 OF 35 / PURE RAW EFFECT

7,200 pieces	1 sticker per piece	Local group $\{1\}$	142 stored moves

$$W_{19} = \mathcal{C}(V^4; T_{14}, H_{56}, H_{29}).$$

Use $U = [H_0, T_1]$, $V = [H_0, T_5]$, and $Z = [[U^4, T_{18}], H_0]$.

Target effect $(P_{3324}\ P_{24987}\ P_{36046})$, with exactly 1 sticker three-cycle. One is $(3843\ 33743\ 49310)$. All other pieces in O19 are fixed.

Piece ID	Hosting cells $H(P)$	Cap signature $M(P)$
3324	$\{8\}$	$\{0, 4, 5, 8, 10, 13, 14, 24, 29, 32, 41\}$
24987	$\{77\}$	$\{4, 24, 29, 64, 68, 70, 77, 82, 101, 111, 134\}$
36046	$\{113\}$	$\{26, 50, 56, 60, 80, 99, 107, 113, 126, 136, 159\}$

Relocate to the two-buffer controller

$$R_{19} = H_{82}\, T_{222}\, T_{404}\, H_{554}\, H_{599}$$
$$(A, B, C) = (3324, 173823, 36046).$$

The reference star has 152 moves. Setup depths range from 0 to 6; all 7,198 required oriented nonbuffer states are reached. A node of depth d costs $152 + 2d$ moves.

Orientation and endgame There is no independent local orientation. Once a legally tracked piece is in its correct position, its sticker arrangement is determined. Use the position insertion routine; do not add a flip algorithm.

Complete support Exactly three target pieces and 3 target stickers move. Every other orbit is fixed at completion.

Native primitive dictionary Positive moves; invert order-three tokens by replacing angle 1 with 2, or 2 with 1.

H_0 = `0:0:1:1`
T_5 = `5:1:1:1`
T_{14} = `14:1:1:1`
H_{29} = `29:0:1:1`
H_{56} = `56:0:1:1`
H_{82} = `82:0:1:1`
T_{222} = `222:1:1:1`
T_{404} = `209:3:1:4`
H_{554} = `43:0:1:4`
H_{599} = `13:2:1:4`

Exact bodies: `O19_exact_words.txt`. Full frames/effects: `seed_atlas.json`. Seed word hash: `4dbdf4044be06aa06eb8`. [S3,S4]

O20 / Rank 10

TARGET SEED / STAGE 29 OF 35 / PURE RAW EFFECT

7,200 pieces	2 stickers per piece	Local group $\{1\}$	142 stored moves

$$W_{20} = \mathcal{C}(U^4; T_{18}, T_{14}, H_{37}).$$

Use $U = [H_0, T_1]$, $V = [H_0, T_5]$, and $Z = [[U^4, T_{18}], H_0]$.

Target effect $(P_{2969}\ P_{18362}\ P_{13799})$, with exactly 2 sticker three-cycles. One is $(3421\ 43194\ 44556)$. All other pieces in O20 are fixed.

Piece ID	Hosting cells $H(P)$	Cap signature $M(P)$
2969	$\{7, 13\}$	$\{0, 1, 3, 6, 7, 9, 11, 13, 16, 17, 37\}$
18362	$\{56, 99\}$	$\{6, 17, 33, 37, 56, 60, 88, 95, 99, 113, 136\}$
13799	$\{41, 102\}$	$\{10, 14, 32, 41, 54, 65, 86, 97, 102, 114, 150\}$

Relocate to the two-buffer controller

$$R_{20} = H_{60}\,H_{244}\,H_{393}\,H_{535}\,H_{595}$$
$$(A, B, C) = (2969, 176860, 13799).$$

The reference star has 152 moves. Setup depths range from 0 to 6; all 7,198 required oriented nonbuffer states are reached. A node of depth d costs $152 + 2d$ moves.

Orientation and endgame There is no independent local orientation. Once a legally tracked piece is in its correct position, its sticker arrangement is determined. Use the position insertion routine; do not add a flip algorithm.

Complete support Exactly three target pieces and 6 target stickers move. Every other orbit is fixed at completion.

Native primitive dictionary Positive moves; invert order-three tokens by replacing angle 1 with 2, or 2 with 1.

H_0 = `0:0:1:1`
T_1 = `1:1:1:1`
T_{14} = `14:1:1:1`
T_{18} = `18:1:1:1`
H_{37} = `37:0:1:1`
H_{60} = `60:0:1:1`
H_{244} = `244:0:1:1`
H_{393} = `218:0:1:4`
H_{535} = `69:6:1:4`
H_{595} = `1:0:1:4`

Exact bodies: `O20_exact_words.txt`. Full frames/effects: `seed_atlas.json`. Seed word hash: `56f951648ab1ca6f1958`. [S3,S4]

O21 / Rank 11

TARGET SEED / STAGE 07 OF 35 / COLLATERAL REQUIRES STAGE ORDER

7,200 pieces	1 sticker per piece	Local group $\{1\}$	94 stored moves

$$W_{21} = \mathcal{C}(V; T_{31}, T_{73}, H_{35}, H_{44}).$$

Use $U = [H_0, T_1]$, $V = [H_0, T_5]$, and $Z = [[U^4, T_{18}], H_0]$.

Target effect $(P_{1129}\ P_{28865}\ P_{18595})$, with exactly 1 sticker three-cycle. One is $(1202\ 39057\ 24563)$. All other pieces in O21 are fixed.

Piece ID	Hosting cells $H(P)$	Cap signature $M(P)$
1129	$\{2\}$	$\{2, 4, 5, 7, 9, 11, 13, 22, 28, 31, 36, 44\}$
28865	$\{90\}$	$\{22, 36, 44, 47, 59, 90, 93, 129, 141, 146, 163, 186\}$
18595	$\{56\}$	$\{12, 17, 33, 35, 56, 67, 88, 99, 133, 137, 152, 183\}$

Relocate to the two-buffer controller

$$R_{21} = T_{93}\,T_{217}\,H_{474}\,T_{560}\,H_{599}$$
$$(A, B, C) = (1129, 175292, 18595).$$

The reference star has 104 moves. Setup depths range from 0 to 6; all 7,198 required oriented nonbuffer states are reached. A node of depth d costs $104 + 2d$ moves.

Orientation and endgame There is no independent local orientation. Once a legally tracked piece is in its correct position, its sticker arrangement is determined. Use the position insertion routine; do not add a flip algorithm.

Complete collateral The full seed moves 27 stickers. In addition to the three target pieces, the following orbits are affected. Entries are orbit / pieces / stickers.

O23 / 3 / 6
O25 / 3 / 6
O26 / 3 / 3
O27 / 3 / 3
O30 / 3 / 6

Native primitive dictionary Positive moves; invert order-three tokens by replacing angle 1 with 2, or 2 with 1.

H_0 = `0:0:1:1`	H_{44} = `44:0:1:1`	H_{474} = `133:6:1:4`
T_5 = `5:1:1:1`	T_{73} = `73:1:1:1`	T_{560} = `35:5:1:4`
T_{31} = `31:1:1:1`	T_{93} = `93:1:1:1`	H_{599} = `13:2:1:4`
H_{35} = `35:0:1:1`	T_{217} = `217:1:1:1`	

Exact bodies: `O21_exact_words.txt`. Full frames/effects: `seed_atlas.json`. Seed word hash: `0f00fc05ed91b00efba1`. [S3,S4]

O22 / Rank 11

TARGET SEED / STAGE 06 OF 35 / COLLATERAL REQUIRES STAGE ORDER

3,600 pieces	2 stickers per piece	Local group C_2	766 stored moves

$$W_{22} = \mathcal{C}(U; H_{15}, H_{49}, H_{53}, T_{55}, T_{76}, T_{63}, H_{69}).$$

Use $U = [H_0, T_1]$, $V = [H_0, T_5]$, and $Z = [[U^4, T_{18}], H_0]$.

Target effect $(P_{5380}\ P_{20589}\ P_{16043})$, with exactly 2 sticker three-cycles. One is $(6693\ 47577\ 21097)$. All other pieces in O22 are fixed.

Piece ID	Hosting cells $H(P)$	Cap signature $M(P)$
5380	$\{15, 23\}$	$\{15, 19, 23, 24, 48, 52, 57, 62, 69, 76, 119, 127\}$
20589	$\{63, 109\}$	$\{16, 20, 47, 48, 52, 53, 55, 62, 63, 66, 85, 109\}$
16043	$\{48, 76\}$	$\{3, 15, 20, 23, 48, 53, 69, 76, 81, 83, 110, 112\}$

Relocate to the two-buffer controller

$$R_{22} = H_{16}\, T_{16}\, T_{80}\, H_{272}\, H_{490}$$
$$(A, B, C) = (5380, 171586, 16043).$$

The reference star has 776 moves. Setup depths range from 0 to 6; all 7,196 required oriented nonbuffer states are reached. A node of depth d costs $776 + 2d$ moves.

Orientation and endgame Place all pieces, then correct flips in pairs with the two-star transfer. Total flip parity is even within this orbit. After the other pieces and buffer A are oriented, buffer B must be correct.

Complete collateral The full seed moves 90 stickers. In addition to the three target pieces, the following orbits are affected. Entries are orbit / pieces / stickers.

- O23 / 6 / 12
- O24 / 6 / 6
- O25 / 3 / 6
- O26 / 3 / 3
- O27 / 6 / 6
- O28 / 3 / 15
- O30 / 12 / 24
- O31 / 6 / 6
- O32 / 3 / 6

Native primitive dictionary Positive moves; invert order-three tokens by replacing angle 1 with 2, or 2 with 1.

- H_0 = `0:0:1:1`
- T_1 = `1:1:1:1`
- H_{15} = `15:0:1:1`
- H_{16} = `16:0:1:1`
- T_{16} = `16:1:1:1`
- H_{49} = `49:0:1:1`
- H_{53} = `53:0:1:1`
- T_{55} = `55:1:1:1`
- T_{63} = `63:1:1:1`
- H_{69} = `69:0:1:1`
- T_{76} = `76:1:1:1`
- T_{80} = `80:1:1:1`
- H_{272} = `272:0:1:1`
- H_{490} = `119:0:1:4`

Exact bodies: `O22_exact_words.txt`. Full frames/effects: `seed_atlas.json`. Seed word hash: `01c71a3e737d0a64419d`. [S3,S4]

O23 / Rank 12

TARGET SEED / STAGE 09 OF 35 / COLLATERAL REQUIRES STAGE ORDER

7,200 pieces	2 stickers per piece	Local group $\{1\}$	46 stored moves

$$W_{23} = \mathcal{C}(V; T_{31}, T_{73}, H_{38}).$$

Use $U = [H_0, T_1]$, $V = [H_0, T_5]$, and $Z = [[U^4, T_{18}], H_0]$.

Target effect $(P_{215}\ P_{1130}\ P_{4596})$, with exactly 2 sticker three-cycles. One is $(215\ 9940\ 5525)$. All other pieces in O23 are fixed.

Piece ID	Hosting cells $H(P)$	Cap signature $M(P)$
215	$\{0, 1\}$	$\{0, 1, 10, 12, 14, 18, 21, 27, 35, 38, 46, 51, 72\}$
1130	$\{2, 22\}$	$\{2, 4, 5, 7, 9, 11, 13, 22, 28, 31, 36, 44, 59\}$
4596	$\{12, 17\}$	$\{0, 5, 6, 10, 12, 13, 17, 26, 35, 38, 56, 60, 67\}$

Relocate to the two-buffer controller

$$R_{23} = H_2\, T_{31}\, H_{199}\, T_{347}\, H_{546}$$
$$(A, B, C) = (215, 176184, 4596).$$

The reference star has 56 moves. Setup depths range from 0 to 6; all 7,198 required oriented nonbuffer states are reached. A node of depth d costs $56 + 2d$ moves.

Orientation and endgame There is no independent local orientation. Once a legally tracked piece is in its correct position, its sticker arrangement is determined. Use the position insertion routine; do not add a flip algorithm.

Complete collateral The full seed moves 24 stickers. In addition to the three target pieces, the following orbits are affected. Entries are orbit / pieces / stickers.

O25 / 3 / 6
O26 / 3 / 3
O27 / 3 / 3
O30 / 3 / 6

Native primitive dictionary Positive moves; invert order-three tokens by replacing angle 1 with 2, or 2 with 1.

H_0 = `0:0:1:1`	T_{31} = `31:1:1:1`	H_{199} = `199:0:1:1`
H_2 = `2:0:1:1`	H_{38} = `38:0:1:1`	T_{347} = `269:3:1:4`
T_5 = `5:1:1:1`	T_{73} = `73:1:1:1`	H_{546} = `51:6:1:4`

Exact bodies: `O23_exact_words.txt`. Full frames/effects: `seed_atlas.json`. Seed word hash: `5d226a531c5401e11186`. [S3,S4]

O24 / Rank 13

TARGET SEED / STAGE 19 OF 35 / COLLATERAL REQUIRES STAGE ORDER

7,200 pieces	1 sticker per piece	Local group $\{1\}$	46 stored moves

$$W_{24} = \mathcal{C}(U; T_{91}, H_7, H_{45}).$$

Use $U = [H_0, T_1]$, $V = [H_0, T_5]$, and $Z = [[U^4, T_{18}], H_0]$.

Target effect $(P_{3298}\ P_{53374}\ P_{5509})$, with exactly 1 sticker three-cycle. One is $(3807\ 74020\ 6846)$. All other pieces in O24 are fixed.

Piece ID	Hosting cells $H(P)$	Cap signature $M(P)$
3298	$\{8\}$	$\{0, 5, 6, 8, 10, 13, 26, 30, 38, 39, 45, 49, 74, 91\}$
53374	$\{170\}$	$\{38, 45, 65, 91, 106, 124, 170, 178, 179, 187, 189, 195, 227, 239\}$
5509	$\{15\}$	$\{2, 3, 7, 11, 15, 16, 19, 20, 22, 47, 48, 52, 53, 62\}$

Relocate to the two-buffer controller

$$R_{24} = T_{65}\, H_{142}\, T_{285}\, H_{424}\, H_{579}$$
$$(A, B, C) = (3298, 173734, 5509).$$

The reference star has 56 moves. Setup depths range from 0 to 6; all 7,198 required oriented nonbuffer states are reached. A node of depth d costs $56 + 2d$ moves.

Orientation and endgame There is no independent local orientation. Once a legally tracked piece is in its correct position, its sticker arrangement is determined. Use the position insertion routine; do not add a flip algorithm.

Complete collateral The full seed moves 6 stickers. In addition to the three target pieces, the following orbits are affected. Entries are orbit / pieces / stickers.

O27 / 3 / 3

Native primitive dictionary Positive moves; invert order-three tokens by replacing angle 1 with 2, or 2 with 1.

H_0 = `0:0:1:1`
T_1 = `1:1:1:1`
H_7 = `7:0:1:1`
H_{45} = `45:0:1:1`
T_{65} = `65:1:1:1`
T_{91} = `91:1:1:1`
H_{142} = `142:0:1:1`
T_{285} = `284:1:1:1`
H_{424} = `212:0:1:4`
H_{579} = `26:0:1:4`

Exact bodies: `O24_exact_words.txt`. Full frames/effects: `seed_atlas.json`. Seed word hash: `3fd0562e3c0072de16ba`. [S3,S4]

O25 / Rank 13

TARGET SEED / STAGE 12 OF 35 / COLLATERAL REQUIRES STAGE ORDER

3,600 pieces	2 stickers per piece	Local group C_2	46 stored moves

$$W_{25} = \mathcal{C}(V; T_{31}, T_{73}, H_{91}).$$

Use $U = [H_0, T_1]$, $V = [H_0, T_5]$, and $Z = [[U^4, T_{18}], H_0]$.

Target effect $(P_{1131}\ P_{4597}\ P_{35851})$, with exactly 2 sticker three-cycles. One is $(1204\ 7709\ 49018)$. All other pieces in O25 are fixed.

Piece ID	Hosting cells $H(P)$	Cap signature $M(P)$
1131	$\{2, 22\}$	$\{2, 4, 5, 7, 9, 11, 13, 22, 28, 31, 34, 36, 44, 59\}$
4597	$\{12, 17\}$	$\{0, 5, 6, 10, 12, 13, 17, 26, 35, 38, 56, 60, 67, 91\}$
35851	$\{113, 166\}$	$\{56, 60, 74, 91, 99, 113, 136, 154, 166, 188, 232, 237, 244, 277\}$

Relocate to the two-buffer controller

$$R_{25} = H_6\, H_{27}\, H_{176}\, H_{401}\, T_{536}$$
$$(A, B, C) = (1131, 170089, 35851).$$

The reference star has 56 moves. Setup depths range from 0 to 6; all 7,196 required oriented nonbuffer states are reached. A node of depth d costs $56 + 2d$ moves.

Orientation and endgame Place all pieces, then correct flips in pairs with the two-star transfer. Total flip parity is even within this orbit. After the other pieces and buffer A are oriented, buffer B must be correct.

Complete collateral The full seed moves 18 stickers. In addition to the three target pieces, the following orbits are affected. Entries are orbit / pieces / stickers.

O26 / 3 / 3
O27 / 3 / 3
O30 / 3 / 6

Native primitive dictionary Positive moves; invert order-three tokens by replacing angle 1 with 2, or 2 with 1.

H_0 = `0:0:1:1`
T_5 = `5:1:1:1`
H_6 = `6:0:1:1`
H_{27} = `27:0:1:1`
T_{31} = `31:1:1:1`
T_{73} = `73:1:1:1`
H_{91} = `91:0:1:1`
H_{176} = `176:0:1:1`
H_{401} = `182:6:1:4`
T_{536} = `59:5:1:4`

Exact bodies: `O25_exact_words.txt`. Full frames/effects: `seed_atlas.json`. Seed word hash: `ea0ab987b2e777ce1444`. [S3,S4]

O26 / Rank 14

TARGET SEED / STAGE 18 OF 35 / COLLATERAL REQUIRES STAGE ORDER

7,200 pieces	1 sticker per piece	Local group $\{1\}$	46 stored moves

$$W_{26} = \mathcal{C}(V; T_{31}, T_{73}, H_{74}).$$

Use $U = [H_0, T_1]$, $V = [H_0, T_5]$, and $Z = [[U^4, T_{18}], H_0]$.

Target effect $(P_{1132}\ P_{6175}\ P_{70620})$, with exactly 1 sticker three-cycle. One is $(1205\ 7710\ 99144)$. All other pieces in O26 are fixed.

Piece ID	Hosting cells $H(P)$	Cap signature $M(P)$
1132	$\{2\}$	$\{2, 4, 5, 7, 9, 11, 13, 22, 28, 30, 31, 34, 36, 44, 59\}$
6175	$\{17\}$	$\{0, 5, 6, 10, 12, 13, 17, 26, 35, 38, 56, 60, 67, 74, 91\}$
70620	$\{228\}$	$\{49, 74, 79, 100, 118, 125, 154, 170, 171, 188, 206, 207, 221, 228, 268\}$

Relocate to the two-buffer controller

$$R_{26} = T_{10}\, H_{61}\, H_{213}\, H_{397}\, H_{571}$$
$$(A, B, C) = (1132, 175303, 70620).$$

The reference star has 56 moves. Setup depths range from 0 to 6; all 7,198 required oriented nonbuffer states are reached. A node of depth d costs $56 + 2d$ moves.

Orientation and endgame There is no independent local orientation. Once a legally tracked piece is in its correct position, its sticker arrangement is determined. Use the position insertion routine; do not add a flip algorithm.

Complete collateral The full seed moves 9 stickers. In addition to the three target pieces, the following orbits are affected. Entries are orbit / pieces / stickers.

O30 / 3 / 6

Native primitive dictionary Positive moves; invert order-three tokens by replacing angle 1 with 2, or 2 with 1.

H_0 = `0:0:1:1`
T_5 = `5:1:1:1`
T_{10} = `10:1:1:1`
T_{31} = `31:1:1:1`
H_{61} = `61:0:1:1`
T_{73} = `73:1:1:1`
H_{74} = `74:0:1:1`
H_{213} = `213:0:1:1`
H_{397} = `186:0:1:4`
H_{571} = `28:0:1:4`

Exact bodies: `026_exact_words.txt`. Full frames/effects: `seed_atlas.json`. Seed word hash: `ca46b93f6188d9b8b2ea`. [S3,S4]

O27 / Rank 14

TARGET SEED / STAGE 28 OF 35 / PURE RAW EFFECT

7,200 pieces	1 sticker per piece	Local group $\{1\}$	94 stored moves

$$W_{27} = [W_{24}, H_{35}].$$

Use $U = [H_0, T_1]$, $V = [H_0, T_5]$, and $Z = [[U^4, T_{18}], H_0]$.

Target effect $(P_{3299}\ P_{50961}\ P_{5511})$, with exactly 1 sticker three-cycle. One is $(3809\ 70562\ 6848)$. All other pieces in O27 are fixed.

Piece ID	Hosting cells $H(P)$	Cap signature $M(P)$
3299	$\{8\}$	$\{0, 5, 6, 8, 10, 13, 26, 30, 35, 38, 39, 45, 49, 74, 91\}$
50961	$\{162\}$	$\{27, 35, 43, 72, 75, 88, 99, 104, 116, 122, 133, 137, 152, 162, 183\}$
5511	$\{15\}$	$\{2, 3, 7, 11, 15, 16, 19, 20, 22, 47, 48, 52, 53, 62, 63\}$

Relocate to the two-buffer controller

$$R_{27} = T_{27}\, H_{78}\, H_{204}\, H_{413}\, H_{539}$$
$$(A, B, C) = (3299, 173737, 5511).$$

The reference star has 104 moves. Setup depths range from 0 to 6; all 7,198 required oriented nonbuffer states are reached. A node of depth d costs $104 + 2d$ moves.

Orientation and endgame There is no independent local orientation. Once a legally tracked piece is in its correct position, its sticker arrangement is determined. Use the position insertion routine; do not add a flip algorithm.

Complete support Exactly three target pieces and 3 target stickers move. Every other orbit is fixed at completion.

Native primitive dictionary Positive moves; invert order-three tokens by replacing angle 1 with 2, or 2 with 1.

H_0 = `0:0:1:1`	H_{35} = `35:0:1:1`	H_{204} = `204:0:1:1`
T_1 = `1:1:1:1`	H_{45} = `45:0:1:1`	H_{413} = `188:0:1:4`
H_7 = `7:0:1:1`	H_{78} = `78:0:1:1`	H_{539} = `74:0:1:4`
T_{27} = `27:1:1:1`	T_{91} = `91:1:1:1`	

Exact bodies: `027_exact_words.txt`. Full frames/effects: `seed_atlas.json`. Seed word hash: `28a6505ccc6d78b4780b`. [S3,S4]

O28 / Rank 14

TARGET SEED / STAGE 17 OF 35 / COLLATERAL REQUIRES STAGE ORDER

1,440 pieces	5 stickers per piece	Local group C_5	3,070 stored moves

$$W_{28} = \mathcal{C}(W_{22}; H_{51}, H_{19}).$$

Use $U = [H_0, T_1]$, $V = [H_0, T_5]$, and $Z = [[U^4, T_{18}], H_0]$.

Target effect $(P_{1910}\ P_{23564}\ P_{16100})$, with exactly 5 sticker three-cycles. One is $(2120\ 36783\ 21172)$. All other pieces in O28 are fixed.

Piece ID	Hosting cells $H(P)$	Cap signature $M(P)$
1910	$\{4, 5, 8, 29, 32\}$	$\{0, 2, 4, 5, 7, 8, 10, 13, 14, 19, 24, 29, 32, 41, 57\}$
23564	$\{72, 75, 84, 92, 108\}$	$\{12, 21, 27, 35, 38, 46, 51, 54, 61, 65, 72, 75, 84, 92, 108\}$
16100	$\{48, 53, 62, 63, 109\}$	$\{3, 15, 16, 19, 20, 47, 48, 52, 53, 55, 62, 63, 66, 85, 109\}$

Relocate to the two-buffer controller

$$R_{28} = H_{12}\, H_{42}\, H_{226}\, H_{426}\, H_{575}$$
$$(A, B, C) = (1910, 170354, 16100).$$

The reference star has 3,080 moves. Setup depths range from 0 to 6; all 7,190 required oriented nonbuffer states are reached. A node of depth d costs $3,080 + 2d$ moves.

Orientation and endgame Place all pieces, then transfer opposite cyclic twists between a target and buffer B. Use the transported five-way frames. The total twist is zero modulo 5 within this orbit; the final buffer needs no residual correction.

Complete collateral The full seed moves 51 stickers. In addition to the three target pieces, the following orbits are affected. Entries are orbit / pieces / stickers.

O30 / 12 / 24
O31 / 6 / 6
O32 / 3 / 6

Native primitive dictionary Positive moves; invert order-three tokens by replacing angle 1 with 2, or 2 with 1.

H_0 = `0:0:1:1`	H_{49} = `49:0:1:1`	T_{76} = `76:1:1:1`
T_1 = `1:1:1:1`	H_{51} = `51:0:1:1`	H_{226} = `226:0:1:1`
H_{12} = `12:0:1:1`	H_{53} = `53:0:1:1`	H_{426} = `196:0:1:4`
H_{15} = `15:0:1:1`	T_{55} = `55:1:1:1`	H_{575} = `32:0:1:4`
H_{19} = `19:0:1:1`	T_{63} = `63:1:1:1`	
H_{42} = `42:0:1:1`	H_{69} = `69:0:1:1`	

Exact bodies: `O28_exact_words.txt`. Full frames/effects: `seed_atlas.json`. Seed word hash: `e776888ac84a075aadc7`. [S3,S4]

O29 / Rank 15

SPECIAL TARGET SEED / STAGE 11 OF 35 / COLLATERAL REQUIRES STAGE ORDER

2,400 pieces	1 sticker per piece	Local group $\{1\}$	9,214 stored moves

$$W_{29} = \mathcal{C}(U^4; T_{18}, T_{17}, H_{10}, T_{37}, H_{38}, H_{58}, H_{45}, T_{189}, H_{190}).$$

Use $U = [H_0, T_1]$, $V = [H_0, T_5]$, and $Z = [[U^4, T_{18}], H_0]$.

Target effect $(P_{11817}\, P_{14344}\, P_{65180})$, with exactly 1 sticker three-cycle. One is $(15382\ 18846\ 90851)$. All other pieces in O29 are fixed.

Piece ID	Hosting cells $H(P)$	Cap signature $M(P)$
11817	$\{35\}$	$\{35, 38, 45, 65, 67, 75, 91, 92, 106, 124, 133, 137, 178, 179, 187, 189\}$
14344	$\{43\}$	$\{33, 37, 42, 43, 58, 87, 88, 95, 103, 104, 116, 128, 139, 140, 190, 219\}$
65180	$\{209\}$	$\{104, 116, 122, 140, 162, 176, 190, 209, 219, 260, 267, 311, 317, 331, 338, 384\}$

Relocate to the two-buffer controller

$$R_{29} = H_{103}\, H_{234}\, H_{471}$$
$$(A, B, C) = (11817, 166897, 65180).$$

The reference star has 9,220 moves. Setup depths range from 0 to 5; all 2,398 required oriented nonbuffer states are reached. A node of depth d costs $9,220 + 2d$ moves.

Orientation and endgame There is no independent local orientation. Once a legally tracked piece is in its correct position, its sticker arrangement is determined. Use the position insertion routine; do not add a flip algorithm.

Complete collateral The full seed moves 38 stickers. In addition to the three target pieces, the following orbits are affected. Entries are orbit / pieces / stickers.

- O24 / 3 / 3
- O26 / 6 / 6
- O27 / 3 / 3
- O31 / 5 / 5
- O32 / 6 / 12
- O33 / 6 / 6

Native primitive dictionary Positive moves; invert order-three tokens by replacing angle 1 with 2, or 2 with 1.

H_0 = `0:0:1:1`	T_{37} = `37:1:1:1`	T_{189} = `189:1:1:1`
T_1 = `1:1:1:1`	H_{38} = `38:0:1:1`	H_{190} = `190:0:1:1`
H_{10} = `10:0:1:1`	H_{45} = `45:0:1:1`	H_{234} = `234:0:1:1`
T_{17} = `17:1:1:1`	H_{58} = `58:0:1:1`	H_{471} = `106:0:1:4`
T_{18} = `18:1:1:1`	H_{103} = `103:0:1:1`	

Exact bodies: `029_exact_words.txt`. Full frames/effects: `seed_atlas.json`. Seed word hash: `b7d3cbd511192bc20749`. [S3,S4]

O30 / Rank 15

TARGET SEED / STAGE 27 OF 35 / PURE RAW EFFECT

7,200 pieces	2 stickers per piece	Local group $\{1\}$	94 stored moves

$$W_{30} = [W_{26}, H_{73}].$$

Use $U = [H_0, T_1]$, $V = [H_0, T_5]$, and $Z = [[U^4, T_{18}], H_0]$.

Target effect $(P_{1133}\ P_{40038}\ P_{64683})$, with exactly 2 sticker three-cycles. One is $(1206\ 90491\ 99151)$. All other pieces in O30 are fixed.

Piece ID	Hosting cells $H(P)$	Cap signature $M(P)$
1133	$\{2, 22\}$	$\{2, 4, 5, 7, 9, 11, 13, 22, 28, 30, 31, 34, 36, 44, 59, 73\}$
40038	$\{126, 208\}$	$\{40, 50, 73, 80, 105, 107, 117, 126, 153, 164, 165, 205, 208, 220, 229, 245\}$
64683	$\{207, 228\}$	$\{39, 49, 74, 79, 100, 118, 125, 154, 170, 171, 188, 206, 207, 221, 228, 268\}$

Relocate to the two-buffer controller

$$R_{30} = H_{107}\, H_{208}\, H_{452}\, T_{566}\, T_{592}$$
$$(A, B, C) = (1133, 170094, 64683).$$

The reference star has 104 moves. Setup depths range from 0 to 5; all 7,198 required oriented nonbuffer states are reached. A node of depth d costs $104 + 2d$ moves.

Orientation and endgame There is no independent local orientation. Once a legally tracked piece is in its correct position, its sticker arrangement is determined. Use the position insertion routine; do not add a flip algorithm.

Complete support Exactly three target pieces and 6 target stickers move. Every other orbit is fixed at completion.

Native primitive dictionary Positive moves; invert order-three tokens by replacing angle 1 with 2, or 2 with 1.

H_0 = `0:0:1:1`
T_5 = `5:1:1:1`
T_{31} = `31:1:1:1`
H_{73} = `73:0:1:1`
T_{73} = `73:1:1:1`
H_{74} = `74:0:1:1`
H_{107} = `107:0:1:1`
H_{208} = `208:0:1:1`
H_{452} = `147:0:1:4`
T_{566} = `57:3:1:4`
T_{592} = `2:5:1:4`

Exact bodies: `030_exact_words.txt`. Full frames/effects: `seed_atlas.json`. Seed word hash: `0e4f0b66f94272dcc941`. [S3,S4]

O31 / Rank 16

TARGET SEED / STAGE 26 OF 35 / PURE RAW EFFECT

7,200 pieces	1 sticker per piece	Local group $\{1\}$	142 stored moves

$$W_{31} = \mathcal{C}(V^4; T_{14}, H_{56}, H_{11}).$$

Use $U = [H_0, T_1]$, $V = [H_0, T_5]$, and $Z = [[U^4, T_{18}], H_0]$.

Target effect $(P_{2520}\ P_{12266}\ P_{19145})$, with exactly 1 sticker three-cycle. One is $(2832\ 16010\ 25542)$. All other pieces in O31 are fixed.

Piece ID	Hosting cells $H(P)$	Cap signature $M(P)$
2520	$\{6\}$	$\{6, 9, 17, 26, 33, 37, 40, 50, 56, 60, 80, 87, 88, 95, 99, 107, 113\}$
12266	$\{36\}$	$\{2, 4, 5, 7, 9, 11, 22, 28, 30, 31, 34, 36, 40, 44, 50, 59, 73\}$
19145	$\{58\}$	$\{1, 3, 7, 9, 11, 12, 16, 17, 20, 21, 25, 27, 33, 37, 42, 43, 58\}$

Relocate to the two-buffer controller

$$R_{31} = H_2\, T_{19}\, H_{148}\, H_{278}\, H_{500}$$
$$(A, B, C) = (2520, 175707, 19145).$$

The reference star has 152 moves. Setup depths range from 0 to 6; all 7,198 required oriented nonbuffer states are reached. A node of depth d costs $152 + 2d$ moves.

Orientation and endgame There is no independent local orientation. Once a legally tracked piece is in its correct position, its sticker arrangement is determined. Use the position insertion routine; do not add a flip algorithm.

Complete support Exactly three target pieces and 3 target stickers move. Every other orbit is fixed at completion.

Native primitive dictionary Positive moves; invert order-three tokens by replacing angle 1 with 2, or 2 with 1.

H_0 = `0:0:1:1`
H_2 = `2:0:1:1`
T_5 = `5:1:1:1`
H_{11} = `11:0:1:1`
T_{14} = `14:1:1:1`
T_{19} = `19:1:1:1`
H_{56} = `56:0:1:1`
H_{148} = `148:0:1:1`
H_{278} = `278:0:1:1`
H_{500} = `107:6:1:4`

Exact bodies: `031_exact_words.txt`. Full frames/effects: `seed_atlas.json`. Seed word hash: `62ecf7bff3b012e5c775`. [S3,S4]

O32 / Rank 17

TARGET SEED / STAGE 25 OF 35 / PURE RAW EFFECT

3,600 pieces	2 stickers per piece	Local group C_2	94 stored moves

$$W_{32} = \mathcal{C}(U; T_{91}, H_7, H_{66}, T_{24}).$$

Use $U = [H_0, T_1]$, $V = [H_0, T_5]$, and $Z = [[U^4, T_{18}], H_0]$.

Target effect $(P_{8474}\ P_{8477}\ P_{41780})$, with exactly 2 sticker three-cycles. One is $(10816\ 10821\ 84365)$. All other pieces in O32 are fixed.

Piece ID	Hosting cells $H(P)$	Cap signature $M(P)$
8474	$\{24, 57\}$	$\{1, 2, 3, 4, 5, 7, 8, 10, 14, 15, 18, 19, 23, 24, 29, 32, 41, 57\}$
8477	$\{24, 29\}$	$\{2, 3, 4, 5, 7, 8, 10, 13, 14, 15, 18, 19, 23, 24, 29, 32, 41, 57\}$
41780	$\{132, 194\}$	$\{55, 63, 66, 85, 98, 109, 132, 144, 151, 158, 194, 201, 214, 234, 236, 248, 255, 289\}$

Relocate to the two-buffer controller

$$R_{32} = H_{13}\ H_9\ T_{55}\ H_{289}\ H_{476}\ H_{576}$$
$$(A, B, C) = (8474, 168341, 14056).$$

The reference star has 106 moves. Setup depths range from 0 to 6; all 7,196 required oriented nonbuffer states are reached. A node of depth d costs $106 + 2d$ moves.

Orientation and endgame Place all pieces, then correct flips in pairs with the two-star transfer. Total flip parity is even within this orbit. After the other pieces and buffer A are oriented, buffer B must be correct.

Complete support Exactly three target pieces and 6 target stickers move. Every other orbit is fixed at completion.

Native primitive dictionary Positive moves; invert order-three tokens by replacing angle 1 with 2, or 2 with 1.

H_0 = `0:0:1:1`	H_{13} = `13:0:1:1`	T_{91} = `91:1:1:1`
T_1 = `1:1:1:1`	T_{24} = `24:1:1:1`	H_{289} = `288:0:1:1`
H_7 = `7:0:1:1`	T_{55} = `55:1:1:1`	H_{476} = `124:0:1:4`
H_9 = `9:0:1:1`	H_{66} = `66:0:1:1`	H_{576} = `14:0:1:4`

Exact bodies: `032_exact_words.txt`. Full frames/effects: `seed_atlas.json`. Seed word hash: `fbc2507fb107cf2975ab`. [S3,S4]

O33 / Rank 18

TARGET SEED / STAGE 24 OF 35 / PURE RAW EFFECT

2,400 pieces	1 sticker per piece	Local group $\{1\}$	46 stored moves

$$W_{33} = \mathcal{C}(U; H_{15}, H_{56}, H_{21}).$$

Use $U = [H_0, T_1]$, $V = [H_0, T_5]$, and $Z = [[U^4, T_{18}], H_0]$.

Target effect $(P_{2712}\ P_{28074}\ P_{35778})$, with exactly 1 sticker three-cycle. One is $(3029\ 38102\ 48927)$. All other pieces in O33 are fixed.

Piece ID	Hosting cells $H(P)$	Cap signature $M(P)$
2712	$\{6\}$	$\{0, 1, 3, 6, 7, 9, 11, 12, 13, 16, 17, 20, 21, 27, 33, 37, 42, 43, 58\}$
28074	$\{87\}$	$\{6, 9, 17, 26, 33, 37, 40, 50, 56, 80, 87, 88, 95, 99, 107, 113, 126, 136, 159\}$
35778	$\{112\}$	$\{3, 15, 18, 20, 21, 23, 25, 46, 48, 51, 53, 69, 71, 76, 78, 81, 83, 110, 112\}$

Relocate to the two-buffer controller

$$R_{33} = H_{26}\, H_{59}\, H_{199}\, H_{428}\, T_{554}$$
$$(A, B, C) = (2712, 175618, 35778).$$

The reference star has 56 moves. Setup depths range from 0 to 5; all 2,398 required oriented nonbuffer states are reached. A node of depth d costs $56 + 2d$ moves.

Orientation and endgame There is no independent local orientation. Once a legally tracked piece is in its correct position, its sticker arrangement is determined. Use the position insertion routine; do not add a flip algorithm.

Complete support Exactly three target pieces and 3 target stickers move. Every other orbit is fixed at completion.

Native primitive dictionary Positive moves; invert order-three tokens by replacing angle 1 with 2, or 2 with 1.

H_0 = `0:0:1:1`
T_1 = `1:1:1:1`
H_{15} = `15:0:1:1`
H_{21} = `21:0:1:1`
H_{26} = `26:0:1:1`
H_{56} = `56:0:1:1`
H_{59} = `59:0:1:1`
H_{199} = `199:0:1:1`
H_{428} = `157:6:1:4`
T_{554} = `43:5:1:4`

Exact bodies: `033_exact_words.txt`. Full frames/effects: `seed_atlas.json`. Seed word hash: `72e65d34d07d48f719f8`. [S3,S4]

O34 / Rank 19

SPECIAL TARGET SEED / STAGE 16 OF 35 / COLLATERAL REQUIRES STAGE ORDER

120 pieces	20 stickers per piece	Local group A_5	12,286 stored moves

$$W_{34} = \mathcal{C}(H_0;H_{24}, H_{18}, H_{45}, H_{27}, H_{99}, H_{38}, H_{21}, H_{156}, H_{54}, H_{147}, H_{352}, H_{233}).$$

Use $U = [H_0, T_1]$, $V = [H_0, T_5]$, and $Z = [[U^4, T_{18}], H_0]$.

Target effect $(P_{5060}\ P_{41186}\ P_{17810})$, with exactly 20 sticker three-cycles. One is $(6300\ 173199\ 53930)$. All other pieces in O34 are fixed.

Piece ID	Hosting cells $H(P)$	Cap signature $M(P)$
5060	$\{14, 18, 23, 41, 51, 54, 57, 61, 69, 81, 97, 102, 119, 127, 130, 138, 142, 147, 173, 184\}$	$\{14, 18, 23, 41, 51, 54, 57, 61, 69, 81, 97, 102, 119, 127, 130, 138, 142, 147, 173, 184\}$
41186	$\{130, 142, 147, 173, 184, 192, 223, 233, 246, 251, 264, 285, 291, 303, 328, 349, 352, 366, 370, 399\}$	$\{130, 142, 147, 173, 184, 192, 223, 233, 246, 251, 264, 285, 291, 303, 328, 349, 352, 366, 370, 399\}$
17810	$\{54, 61, 65, 92, 97, 108, 124, 130, 142, 150, 179, 192, 195, 200, 215, 233, 235, 246, 254, 297\}$	$\{54, 61, 65, 92, 97, 108, 124, 130, 142, 150, 179, 192, 195, 200, 215, 233, 235, 246, 254, 297\}$

Relocate to the two-buffer controller

$$R_{34} = H_{370}\, H_{550}\, H_{594}\, H_{566}$$
$$(A, B, C) = (5060, 130196, 17810).$$

The reference star has 12,294 moves. Setup depths range from 0 to 5; all 7,080 required oriented nonbuffer states are reached. A node of depth d costs $12,294 + 2d$ moves.

Orientation and endgame Use the full set of sixty local frames for the twenty stickers. A single residual corner orientation is legal. The eight-star commutator handles the final buffer; the additional Nan Ma G/H pair is only simplified-subsystem-pure.

Complete collateral The full seed moves 81 stickers. In addition to the three target pieces, the following orbits are affected. Entries are orbit / pieces / stickers.

O33 / 21 / 21

Native primitive dictionary Positive moves; invert order-three tokens by replacing angle 1 with 2, or 2 with 1.

H_0 = `0:0:1:1`
H_{18} = `18:0:1:1`
H_{21} = `21:0:1:1`
H_{24} = `24:0:1:1`
H_{27} = `27:0:1:1`
H_{38} = `38:0:1:1`
H_{45} = `45:0:1:1`
H_{54} = `54:0:1:1`
H_{99} = `99:0:1:1`
H_{147} = `147:0:1:1`
H_{156} = `156:0:1:1`
H_{233} = `233:0:1:1`
H_{352} = `245:0:1:4`
H_{370} = `229:0:1:4`
H_{550} = `50:0:1:4`
H_{566} = `57:2:1:4`
H_{594} = `5:0:1:4`

Exact bodies: `034_exact_words.txt`. Full frames/effects: `seed_atlas.json`. Seed word hash: `f08fd3dfc839e792a7c6`. [S3,S4]

8. The four special target seeds

These are the exact four entries in `target_special_seeds.json`, already included in the 35-card atlas. Their purpose in the solver is the same as every other target seed: provide an order-three transport of three target pieces that can be relocated and conjugated into stars. They are not orientation-only algorithms. [S3,S4]

Orbit	Stored moves	What the complete seed does
O15	11,564	Cycles three rank-9 single-sticker pieces; also affects 22 other orbits.
O17	1,534	Cycles three rank-9 five-sticker pieces; also affects 19 other orbits.
O29	9,214	Cycles three rank-15 single-sticker pieces; also affects six other orbits.
O34	12,286	Cycles three twenty-sticker corners and also moves 21 stickers in O33.

A readable construction of the long O15 seed

Start with $E_0 = T_0$. Repeatedly replace E by $[E, g]$, following this exact ordered list:

$$T_{19}, T_{57}, T_0, T_{24}, T_0, T_{24}, T_0^{-1}, T_{24}, T_0^{-1}, T_{24}, T_0, T_{24}.$$

The result is W_{15}. Literal recursive expansion has 12,286 moves. Reducing adjacent powers with $T_c^3 = 1$ gives the stored 11,564-move word exactly. No geometric approximation is involved in this equivalence check. The appearances of T_0^{-1} matter; replacing them by T_0 changes the algorithm.

A readable construction of the O34 corner seed

Start with $E_0 = H_0$ and successively commutate with

$$H_{24}, H_{18}, H_{45}, H_{27}, H_{99}, H_{38}, H_{21}, H_{156}, H_{54}, H_{147}, H_{352}, H_{233}.$$

This yields the stored 12,286 moves. Only O33 remains as collateral, which is why O34 can be completed before O33 in the production order. Although a four-move corner cycle exists for the simplified subsystem, that shorter word has much broader full-model collateral and is not a drop-in replacement.

Seed refinement versus manual workload

Long seeds inflate the primitive count even when the number of target selections is modest. In the seed-600 run, O15, O29 and O02 account for 14,193 stars but 142,277,598 primitive moves, about 40.34% of the primitive total. Finding shorter words with compatible collateral on those three orbits is a particularly direct optimization target. It would mainly reduce backend execution and proof size; it does not automatically reduce the number of human target-selection decisions.

9. Additional algorithms from Nan Ma's log

The native blocks below are taken from the supplied simplified log. Their IDs denote their ordinal macro blocks in the analysis. They are useful short algorithms, but their full-model collateral must remain visible in any proof or preservation check. The complete original log is included unchanged. It explicitly has `Simplified` and a 9,000-sticker saved field. [S2,S5]

Block	Moves	Simplified-subsystem role	Full stickers affected
1	8	Ridge-piece three-cycle	5,259
13	16	Two ridge flips in place	6,005
5127	4	Edge three-cycle, preserving ridges	3,502
5143	10	Reflection-type orientations on two edges	2,556
5132	24	One stationary edge's five-way rotation	2,271
9358	4	Corner three-cycle, preserving simplified ridges/edges	1,530
9363	10	One stationary corner orientation	975

Four-move corner cycle, block 9358

```
98:3:2:4 25:4:1:4 98:3:1:4 25:4:2:4
```

It cycles three corners in the simplified subsystem. On the full model it affects 1,005 pieces spread over O10 through O34. This is the main reason a visible three-cycle cannot be assumed pure on the full puzzle.

The same-corner generators G and H

```
G = 24:1:1:1 0:3:2:1 57:1:1:1 0:3:1:1 57:1:2:1
    24:1:2:1 57:1:1:1 0:3:2:1 57:1:2:1 0:3:1:1
H = 23:1:1:1 5:5:2:1 18:1:1:1 5:5:1:1 18:1:2:1
    23:1:2:1 18:1:1:1 5:5:2:1 18:1:2:1 5:5:1:1
```

Both operate at the same reference corner, lab piece 432. Their exact local permutations generate all 60 corner orientations. The verified lookup table needs at most six applications from G, G^{-1}, H, H^{-1}, hence at most 60 native moves at that reference corner. For another corner, conjugate the whole selected word once with a suitable setup and use the transported table frame. The length bound is $2|S| + 60$.

> G and H fix the other simplified pieces, but they disturb extra full-model pieces in O10 through O33. They do not replace a pure full-puzzle corner-orientation algorithm after those extra orbits have been solved. The full table and exact frame order are in `CORNER_ORIENTATION_TABLE.json`. [S5]

10. Workload: measured counts and bounds

The three 1,000-move random trials and the cleanup of the full state obtained from Nan Ma's history all use the recorded raw-seed triangular method. The 200-move pure-controller demonstration is a separate, much more expensive primitive-word construction. These samples are not a statistical study of uniformly random states. [S4]

Recorded trace	Stars	Solution primitives
Random seed 600, 1,000-move scramble	374,178	352,698,206
Random seed 601, 1,000-move scramble	373,948	352,080,470
Random seed 602, 1,000-move scramble	374,428	352,839,050
Lifted Nan Ma residual, cleanup only	373,962	346,312,408
Pure-controller trial, 200-move scramble	340,422	624,752,715,364

The complete combined native export for the lifted solve also contains Nan Ma's original 12,000-record scramble and 102,312-record simplified solve. Its full solution portion therefore has 346,414,720 native primitive records, and its entire history has 346,426,720. The cleanup table above excludes that original prefix. The native general twists in the original prefix and the lab primary generators are different counting alphabets; the original 114,312 native records expand to 248,361 lab primitives.

Where the stars are spent in seed 600

Phase	Stars	Share
Piece permutation	345,833	92.425%
Nonbuffer orientation	28,302	7.564%
Buffer A orientation	27	0.007%
Buffer B residual commutators	16	0.004%

A bound for this exact controller scheme

A positional insertion costs at most two stars per nonbuffer piece. A nontrivial local orientation costs at most two more; buffer A costs at most three, and a nonabelian buffer B at most eight. Let $a_o = 1$ for a nontrivial local group and $b_o = 1$ for D_5 or A_5. Then

$$S_o \leq 2(N_o - 2) + a_o\big(2(N_o - 2) + 3 + 8b_o\big).$$

Summing over the 35 orbits gives **405,945 stars** as a conservative upper bound for legal inputs to this controller scheme. It excludes preparation of the controller bank, setup-tree construction and input translation. It is not a lower bound on the puzzle and does not limit better methods. Using each orbit's maximum certified setup depth gives the corresponding conservative primitive bound

$$\sum_o S_o^{\max}(\ell_o + 2r_o + 2d_o^{\max}) = 385,797,152.$$

11. Human time versus machine time

A practical estimate must specify the execution mode. A native macro that still animates or individually applies its thousands of primitive moves has very different cost from an enhanced client applying a certified complete net permutation atomically. Headless solving times reported earlier exclude graphics and do not measure a human solve.

For a human using atomic certified stars, define t_{sel} as target/orientation recognition time, t_{setup} as setup selection time, t_{check} as checking time, and t_{apply} as application latency. For S stars,

$$T_{\mathrm{human}} \approx \frac{S(t_{\mathrm{sel}} + t_{\mathrm{setup}} + t_{\mathrm{check}} + t_{\mathrm{apply}})}{3600} \text{ hours.}$$

The following are transparent throughput scenarios for the recorded 374,178-star trace, not measured human performances. Learning, building tools, breaks, recovery and later optimization are excluded.

Seconds per star	Active hours	At 2 hours/day	Interpretation
1	103.9	52 days	Mostly automatic target selection; very optimistic for manual inspection.
5	519.7	260 days	Highly assisted, fast target recognition.
15	1,559.1	780 days	Manual targeting with reliable lookup and atomic execution.
30	3,118.2	1,559 days	More deliberate recognition, orientation and checking.
60	6,236.3	3,118 days	Slow visual navigation or frequent manual setup work.

Planning interpretation. A 15 to 30 second-per-star scenario means roughly 1,560 to 3,120 active hours with these algorithms. That is a workload model, not a prediction of your speed. Benchmark a representative batch in the intended interface before committing to that range. The 1 to 5 second scenarios require much more automated selection and recognition.

Primitive-by-primitive execution is a different workload

For $L = 352,698,206$ primitive moves and sustained throughput v,

$$T_{\mathrm{primitive}} = L/(3600v) \text{ hours.}$$

Sustained primitives per second	Execution hours only
1	97,971.7
5	19,594.3
10	9,797.2
100	979.7
1,000	98.0

At one primitive per second this is about 11.18 years of nonstop execution. These rates are hypothetical, not measured MPUlt frame rates. The present primitive words are plainly unsuitable for hand entry. Cached macro execution, exact orbit filtering, reliable target lookup and checkpointing are prerequisites for making this route manageable.

Color versus labelled solving. The verified solver returns every tracked sticker identity home, including identical-color copies. That is stronger than visual color completion. A human-oriented method could avoid some identity work, but no reduced human workload or new move bound has been measured here.

12. Workload by orbit, recorded seed 600

Stage	Orbit	Stars	Primitive moves	Mean setup depth	Maximum depth used
1	O06	2,676	1,092,462	5.12	9
2	O00	2,600	5,822,558	10.72	20
3	O17	5,140	7,985,676	3.82	6
4	O15	4,775	55,307,420	4.35	6
5	O02	4,648	42,959,544	6.29	10
6	O22	10,760	8,434,022	3.92	6
7	O21	14,368	1,606,068	3.89	6
8	O08	10,778	16,743,356	4.74	7
9	O23	14,365	920,818	4.05	6
10	O09	14,315	5,773,026	4.64	7
11	O29	4,770	44,010,634	3.27	5
12	O25	10,724	676,082	3.52	6
13	O13	14,345	2,302,110	4.24	6
14	O11	10,822	8,489,924	4.25	6
15	O10	12,972	10,186,616	4.64	7
16	O34	473	5,817,876	2.97	5
17	O28	5,134	15,848,714	3.51	6
18	O26	14,354	907,660	3.62	6
19	O24	14,358	913,914	3.83	6
20	O16	14,232	4,323,956	3.91	6
21	O14	13,382	20,773,482	4.17	6
22	O12	14,151	11,100,498	4.22	6
23	O03	10,547	11,834,790	6.05	10
24	O33	4,779	300,420	3.43	5
25	O32	10,749	1,212,184	3.39	6
26	O31	14,359	2,293,398	3.86	6
27	O30	14,354	1,595,572	3.58	5
28	O27	14,369	1,602,806	3.77	6
29	O20	14,373	2,298,908	3.97	6
30	O19	14,370	2,300,664	4.05	6
31	O18	14,303	22,221,056	3.80	6
32	O07	13,055	4,025,696	5.18	8
33	O05	14,102	8,076,348	5.35	8
34	O04	13,968	8,003,658	5.50	8
35	O01	6,708	14,936,290	7.32	12

This table counts actual stored commands and their exact expansions. Setup depth excludes the seed relocation. The maximum depth used in this particular trace may be smaller than the full-tree maximum on the corresponding algorithm card. [S4]

13. Actual log files and what is verified

The compact records contain real commands

The three random-solve certificates contain the original scramble, the 35 controller seed/relocation definitions, the solve order and every command. A command is $[o, n, \sigma]$: orbit, oriented tree-node index, and direction $\sigma = \pm 1$. It expands deterministically to

$$S_n^{-1} R_o^{-1} W_o^{\sigma} R_o S_n.$$

The cleanup certificate starts from `lifted_final_labels.npy`; its state hash was checked against a fresh replay of all 114,312 native records in Nan Ma's supplied history. The pure-controller trace uses its separate recursive macro certificate. These files are move logs in compressed algorithmic form, not just result summaries. [S2,S4,S5]

File in the reference package	Contents
`triangular_solution_0.json.gz`	Seed 600, 1,000 scramble moves and 374,178 stars.
`verified_solution_601.json.gz`	Seed 601, 1,000 scramble moves and 373,948 stars.
`verified_solution_602.json.gz`	Seed 602, 1,000 scramble moves and 374,428 stars.
`full_residual_cleanup.json.gz`	373,962-star cleanup of the exact lifted residual.
`solution_0.json.gz`	200-move pure-controller trial, with `macro_certificate.json.gz`.
`600cell_solved.log`	Original simplified log, unchanged.

The expanded archival logs contain every native move

The separately supplied `*.native.log.gz` files decompress to complete MPUlt v1-format files: full puzzle definition, 259,800 saved colors, scramble marker, every primitive token, macro boundaries, checksum and final terminator. No controller ID or compressed star command is left in the expanded move text. The stored final colors come from the solved result; opening that color field alone would not constitute a replay proof.

All four exports passed an independent streaming C++ count, syntax and checksum check. Their paths and compressed/uncompressed hashes are in the export reports. That check is deliberately labelled *not a geometry replay*. Geometry correctness is grounded in the full-model command replay plus the verified legal seed/conjugation construction. A primitive-by-primitive replay of all hundreds of millions of moves in Windows was not performed.

Do not feed these multi-GB traces to the original 32-bit program. Seed 600 needs 353,447,562 sequence entries including macro delimiters. Just storing those entries as 64-bit values needs 2.83 GB. The inspected doubling allocator grows to 409,600,000 entries, allocating 3.28 GB; during the last growth the old and new arrays together need 4.92 GB before geometry or graphics memory. A streamed or segmented history design is necessary. [S6]

14. Replaying and inspecting the records

Extract the reference package. Its `engine` subdirectory contains the minimal retained solver/model files needed for replay, so the earlier ZIP is not required. Run commands from the extracted package directory:

```
py -3 -m venv .venv
.venv\Scripts\python.exe -m pip install -r requirements.txt
.venv\Scripts\python.exe replay_logs.py logs/triangular_solution_0.json.gz
.venv\Scripts\python.exe replay_logs.py logs/verified_solution_601.json.gz
.venv\Scripts\python.exe replay_logs.py logs/verified_solution_602.json.gz
.venv\Scripts\python.exe replay_logs.py logs/full_residual_cleanup.json.gz
.venv\Scripts\python.exe replay_logs.py logs/solution_0.json.gz
```

The replay script verifies controller definitions and input/output hashes, executes the saved command list, and checks all 259,800 labels. It does not invoke the solver to generate a replacement solution. A successful replay prints `solved: true`. No displayed color or visibility shortcut is used.

Inspect one actual star before execution

```
.venv\Scripts\python.exe inspect_star.py 33 0 1 -out star33.txt
```

This exports the exact lab and native words for node 0, the O33 reference star, and reports its three piece positions and complete ordered frames. Use a different valid node to target another position/frame. The tool rejects invalid orbit, node or direction values.

Archival export and validation

```
.venv\Scripts\python.exe export_native_logs.py logs/triangular_solution_0.json.gz -output
full_seed600.native.log.gz
```

This writes a complete archival export, not a GUI-generated save. Regenerating the residual export additionally requires `-original-log logs/600cell_solved.log`. Its timer is zero because no native GUI solve time was measured. The format validator can be built in a C++17 environment with zlib:

```
g++ -O3 -std=c++17 validate_native_log.cpp -lz -o validate_native_log
./validate_native_log full_seed600.native.log.gz
```

It reads gzip directly and avoids storing the full primitive history. It checks the file format, legal token ranges, record count, macro nesting, scramble boundary, saved color field and native checksum. Its output explicitly says `geometry_replayed:false`. Native GUI acceptance requires the separate geometry bridge and a history implementation capable of handling the file.

15. Geometry, provenance and evidence limits

The geometry is the exact-golden-ratio reconstruction of the supplied rounded definition. With $\phi = (1+\sqrt{5})/2$, the initial cell pole is $n_0 = (\phi^3, 1, 1, 1)$. The group expansion fixes all lab cell IDs and frames F_c in `cells.json`. Define the reflection

$$\mathcal{R}(v) = I - 2vv^\mathsf{T}/(v^\mathsf{T}v).$$

The two named generators at cell c apply the rotations

$$H_c : F_c\mathcal{R}(v_2)\mathcal{R}(v_1)F_c^\mathsf{T},$$
$$T_c : F_c\mathcal{R}(w_2)\mathcal{R}(w_1)F_c^\mathsf{T},$$

where

$$v_1 = (0, \phi, 1-\phi, -1), \quad v_2 = (1-\phi, 1, 0, \phi),$$
$$w_1 = (0, 2, -1, -1), \quad w_2 = (0, 1, 1, -2).$$

Their support is the cap $n_c \cdot x > 0.968\|n_c\|^2$. Piece signatures use all such caps. For this model, $r(P) = |M(P)| - 1$ and the number of stickers is $|H(P)|$. [S1,S3]

What this edition checked directly

All 35 compact formulas reduce to the stored words; all 35 stored words have their recorded full sticker signatures; all target effects are order-three transports of exactly three pieces; all 35 actual buffer pairs are antipodal. The complete saved command traces were replayed for three random full solves, the lifted residual cleanup and the pure-controller trial. The original native history was replayed to the saved full residual again. The expanded native-format archives passed an independent stream-level syntax/count/checksum test. Exact machine-readable reports are supplied. [S4]

The exhaustive orientation-group and all-primary-generator invariant reports, plus the 70 forward/inverse expanded-star witness checks, are retained from the earlier toolkit. Their scope and filenames are preserved. A full-state solve replay applies computed complete macro permutations rather than rendering or individually replaying every primitive. This distinction matters for both verification claims and workload estimates.

Source register

S1. User-supplied `MPUlt_puzzles(2).txt`, full definition at lines 981 to 989, and the embedded definition in the supplied log. Cuts and allowed layers define the scope of every formula here.

S2. User-supplied `600cell_solved.log`. SHA-256 begins `71b0b1262391db2cd3c8a29a267068b43`. Original simplified history, 12,000 scramble records and 102,312 solution records. No author, camera or selected-target metadata is inferred from absent fields.

S3. Retained `full_600cell_toolkit`: `target_seeds.json`, `target_special_seeds.json`, `buffer_certificates.json`, `census.json`, `model.npz`, and the controller implementation. The O-number vocabulary and raw-seed stage order originate here.

S4. This edition's `audits` directory, especially `reference_audit.json`, `compact_formula_audit.json`, `seed600_phase_workload.json`, `workload.json`, the replay reports, and native-stream checks. Human throughput tables are explicit calculations under stated assumptions, not source-reported solve times.

S5. Retained `600cell_log_analysis`: native translation, log audit, selected macros and the 60-state corner table. The short Nan Ma macro effects and reference corner frames derive from these files.

S6. Public `cutelyaware/MPUlt`, `src/Puzzle.cs`, blob `144e56f05de4c31045e3e0f5c3a19513d6a03618`. The native packing, base-36 color encoding, checksum, saving/loading conventions and doubling history allocator were checked in this source.

The reference package includes exact words, frame trees, original input, replay software, all compact solve records and a hash manifest. Expanded native archives are distributed separately to keep the working package small.